# Solutions for Analysis I by Terence Tao

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## Contents



*Remark.* This document follows the third edition of *Analysis I*.

# 1 Introduction

No exercises from this chapter.

# 2 Starting at the beginning: the natural numbers

## 2.1 The Peano axioms

No exercises from this subchapter.

## 2.2 Addition

**Exercise 2.2.1**. *Prove Proposition 2.2.5 (Associativity of addition): For any natural numbers $a, b, c$, we have $(a + b) + c = a + (b + c)$.*

*Solution.* We induct on $a$, keeping $b$ and $c$ fixed.

First, consider the base case $a = 0$. Then we have $(0 + b) + c = 0 + (b + c)$. By Lemma 2.2.2, we obtain $b + c = b + c$ as desired.

Now we assume inductively that $(a + b) + c = a + (b + c)$; we now have to show $((a{++}) + b) + c = (a{++}) + (b + c)$ to close the induction. By applying Lemma 2.2.3 twice on the left-hand side, $((a{++})+b)+c = ((a+b){++})+c = (a+b+c){++}$. Similarly, applying Lemma 2.2.3 on the right-hand side yields $(a{++}) + (b + c) = (a + b + c){++}$. Thus, both sides are equal, and we have closed the induction. □

**Exercise 2.2.2**. *Prove Lemma 2.2.10: Let $a$ be a positive number. Then there exists exactly one natural number $b$ such that $b{++} = a$.*

*Solution.* We induct on $a$.

First, consider the base case $a = 1$. By Definition 2.1.3, $1 = 0{++}$. Suppose there is another natural number $b$ such that $b{++} = 1$. However, by Axiom 2.4, if $b{++} = 0{++} = 1$, then $b = 0$. Thus, there is exactly one natural number $b$ such that $b{++} = 1$. Hence, the base case is proven.

Suppose inductively that there exists exactly one natural number $b$ such that $b{++} = a$. We now have to show there exists exactly one natural number $b'$ such that $b'{++} = a{++}$. As per Axiom 2.4, if $b'{++} = a'{++}$, then we must have $b' = a$, which is the only solution. This closes the induction. □

**Exercise 2.2.3**. *Prove Proposition 2.2.12 (Basic properties of order of natural numbers).*

*Solution.* Let $a, b, c$ be natural numbers. Then,

(a) *Reflexivity:* $a \geq a$.

By Lemma 2.2.2, $a + 0 = a$. Thus, by Definition 2.2.11, $a \geq a$.

(b) *Transitivity: If* $a \geq b$ *and* $b \geq c$*, then* $a \geq c$.

By Definition 2.2.11, write $a = b + m$, $b = c + n$, where $m$ and $n$ are natural numbers. Thus, $a = (c+n)+m = c+(n+m)$ by associativity of addition. By mathematical induction, it is trivial to prove that $n + m$ is a natural number. Therefore, $a \geq c$.

(c) *Anti-symmetry: If* $a \geq b$ *and* $b \geq a$*, then* $a = b$.

By Definition 2.2.11, if $a \geq b$, then for a natural number $m$, $a = b + m$. But also if $b \geq a$, then $b = a + n$, where $n$ is a natural number. Thus $a = (a+n)+m = a+(n+m)$. By Lemma 2.2.2 and Proposition 2.2.6 (cancellation law), $0 = n + m$. By Corollary 2.2.9, this results in $m = n = 0$. Thus $a = b + 0 = b$ as required.

(d) *Preservation of order:* $a \geq b$ *if and only if* $a + c \geq b + c$.

First, prove $a \geq b \implies a + c \geq b + c$. According to Definition 2.2.11, if $a \geq b$, then $a = b + m$, where $m$ is a natural number. Thus $a + c = (b + m) + c = (b + c) + m$, thus $a + c \geq b + c$.

Conversely, consider $a + c = (b + c) + m$, where $m$ is a natural number. By Proposition 2.2.6 (cancellation law) and the associativity of addition, $a = b+m$, meaning $a \geq b$.

Because $a \geq b \implies a + c \geq b + c$ and $a + c \geq b + c \implies a \geq b$, $a + c \geq b + c \iff a \geq b$.

(e) $a < b$ *if and only if* $a{+}{+} < b$.

First, consider $a < b \implies a{+}{+} \leq b$. According to Definition 2.2.11, if $a < b$, then $b = a + m$, where $m$ is a natural number. Note that $m \neq 0$ because we would otherwise obtain $a = b$, which is contradictory to the given condition $a < b$. Therefore, by Axiom 2.3, we have $m = n{+}{+}$, where $n$ is a natural number. Thus, by the definition and commutativity of addition,

$b = a + n{+\!+} = (a + n){+\!+} = (a{+\!+}) + n$. It follows that $a{+\!+} \leq b$.

Now consider the converse. From $a{+\!+} \leq b$, we obtain $b = a{+\!+} + p$, where $p$ is a natural number. Thus, by the definition and commutativity of addition, $b = a{+\!+} + p = (a + p){+\!+} = a + p{+\!+}$. According to Axiom 2.3, 0 is not the successor of any natural number, so $p{+\!+} \neq 0$. Hence, $a \leq b$.

Because $a < b \implies a{+\!+} \leq b$ and $a{+\!+} \leq b \implies a < b$, $a < b \iff a{+\!+} \leq b$.

(f) *$a < b$ if and only if $b = a + d$ for some positive number $d$.*

First, consider $a < b \implies b = a+d$. From $a < b$ we know $a \geq b$, or $b = a+d$ for a natural number $d$. For the sake of contradiction, suppose $d = 0$. Therefore $a = b$, which contradicts $a < b$. Hence, $d$ is positive; thus, $a < b \implies b = a+d$ for some positive number $d$.

Now consider the converse. From Definition 2.2.11, $b = a + d$ implies $b \geq a$, or $a \leq b$. If $b = a$, by Proposition 2.2.6 (cancellation law), $d = 0$, which contradicts the given condition that $d$ is positive. Therefore, $a \neq b$, which implies $a < b$.

Because the statement and the converse are both true, we can conclude that for some positive number $d$, $a < b$ if and only if $b = a + d$. $\square$

**Exercise 2.2.4**. *Justify the three lemmas marked with (why?) in Proposition 2.2.13 (trichotomy of order for natural numbers).*

*Solution.*

(a) *For all $b$, $0 \leq b$.*

From the definition of addition, $0 + b = b$. By Definition 2.2.11, this implies $0 \leq b$.

(b) *If $a > b$, then $a{+\!+} > b$.*

From Definition 2.2.11, $a = b + m$, where $m$ is a natural number. Thus, by the definition of addition, $a{+\!+} = (b + m){+\!+} = b + m{+\!+}$. Because $m{+\!+}$ is positive by Proposition 2.2.8, $a{+\!+} > b$.

(c) *If $a = b$, then $a{+\!+} > b$.*

From Axiom 2.4, if $a = b$, then $a{+\!+} = b{+\!+}$. Recall that $b{+\!+} = b + 1$, thus $a{+\!+} = b + 1$. By Definition 2.2.11, this implies $a{+\!+} > b$. □

**Exercise 2.2.5**. *Prove Proposition 2.14 (Strong principle of induction): Let $m_0$ be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number $m$. Suppose that for each $m \geq m_0$, we have the following implication: if $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. Then we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.*

*Solution.* Let $Q(n)$ be the property such that $P(m)$ is true for all $m_0 \leq m < n$. Note that the supposed implication becomes "If $Q(n)$ is true, then $P(n)$ is also true. We induct on $n$.

First, consider the base case $n = m_0$. In this case, $Q(m_0)$ is vacuously true as there is no such natural number $m$. Hence, the base case is proven. (Note that $Q(n)$ is also vacuously true for $n < m_0$.)

Now, we assume inductively that $Q(n)$ is true; we will now prove $Q(n{+\!+})$ is true. By the inductive hypothesis, $P(m)$ is true for all $m$ where $m_0 \leq m < n$. According to the supposed implication, because $Q(n)$ is true, $P(n)$ is also true. Therefore, $P(m)$ is true for $m_0 \leq m \leq n$, which implies $Q(n{+\!+})$ is also true. This closes the induction. □

**Exercise 2.2.6**. *Let $n$ be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m{+\!+})$ is true, then $P(m)$ is true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$; this is known as the* principle of backwards induction.

*Solution.* Write $Q(n)$ as the property pertaining to $n$ that "If $P(n)$ is true, $P(m)$ is true for all natural numbers $m \leq n$. We apply induction on $n$.

Consider the base case $n = 0$. The statement $m \leq n$ implies $0 = m + x$ for a natural number $x$. According to Corollary 2.2.9, this leads to $m = x = 0$. Thus, $Q(n)$ is equivalent to "If $P(n)$ is true, $P(n)$ is true.", which is always true regardless of whether $P(n)$ is true or false. Hence, the base case is proven.

Suppose inductively that $Q(n)$ is true; we now need to show $Q(n{+\!+})$ is true. From the definition of $P(m)$, $P(n{+\!+})$ being true implies $P(n)$ is true. By the inductive statement, $P(m)$ is true for all natural numbers $m \leq n$. Thus if $P(n{+\!\!+})$ is true, $P(m)$ is true for all $m \leq n{+\!\!+}$. This is precisely the definition of $Q(n{+\!\!+})$, so $Q(n{+\!\!+})$ is also true. This closes the induction. $\square$

## 2.3 Multiplication

**Exercise 2.3.1**. *Prove Lemma 2.3.2 (Commutativity of multiplication): Let $n, m$ be natural numbers. Then $n \times m = m \times n$.*

*Solution.* First, we introduce a few lemmas. For all lemmas, we induct on $n$, keeping $m$ fixed (when it appears).

**Lemma.** *For any natural number $n$, $n \times 0 = n$.*

*Proof.* In the base case $n = 0$, $0 \times 0 = 0$ is true by the definition of multiplication, and 0 is a natural number. Now suppose inductively that $n \times 0 = 0$; we will now prove $(n{+\!\!+}) \times 0 = 0$. By the definition of multiplication, $(n{+\!\!+}) \times 0 = (n \times 0) + 0 = 0 + 0 = 0$. $\square$

**Lemma.** *For any natural numbers $n$ and $m$, $n \times (m{+\!\!+}) = (n \times m) + n$.*

*Proof.* In the base case $n = 0$, $0 \times m{+\!\!+} = 0$ by the definition of multiplication. Also, $(0 \times m) + 0 = 0$. Thus, the left-hand side is true.

Now suppose inductively that $n \times (m{+\!\!+}) = (n \times m) + n$; we must now show $(n{+\!\!+}) \times (m{+\!\!+}) = ((n{+\!\!+}) \times m) + (n{+\!\!+})$. We start on the left-hand side.

$$\begin{aligned}
&(n{+\!\!+}) \times (m{+\!\!+}) && \\
=&n \times (m{+\!\!+}) + m{+\!\!+} && \text{(Definition 2.3.1)} \\
=&(n \times m) + n + m{+\!\!+} && \text{(Inductive hypothesis)} \\
=&(n \times m) + m + n{+\!\!+} && \text{(Addition properties)} \\
=&((n{+\!\!+}) \times m) + (n{+\!\!+}) && \text{(Definition 2.3.1)}
\end{aligned}$$

This closes the induction. $\square$

We can now prove the main statement $n \times m = m \times n$. Consider the base case $n = 0$. Thus, $0 \times m = m \times 0 = 0$. Hence, the base case is true.

Now suppose inductively that $n \times m = m \times n$; we will now prove $(n{+\!\!+}) \times m = m \times (n{+\!\!+})$. By the definition of multiplication, $(n{+\!\!+}) \times m = (n \times m) + m$. From the second lemma, the right-hand side is also equivalent to $(n \times m) + m$. Because both sides are equal, the statement is true. This closes the induction. $\square$

**Exercise 2.3.2**. *Prove Lemma 2.3.3 (Positive natural numbers have no zero divisors): Let $m, n$ be natural numbers. Then $n \times m = 0$ if and only if at least one of $n, m$ is equal to zero. In particular, if $n$ and $m$ are both positive, then $nm$ is also positive.*

*Solution.* We will prove the second statement first. For the sake of contradiction, suppose there exist positive numbers $m, n$ such that $mn = 0$. Because $m, n$ are positive numbers, write $m = p{+}{+}, n = q{+}{+}$, thus $(q{+}{+}) \times (p{+}{+}) = 0$. From the definition of multiplication, $(q \times (p{+}{+})) + (p{+}{+}) = 0$. According to Corollary 2.2.9, this implies $q \times (p{+}{+}) = p{+}{+} = 0$, which is contradictory as Axiom 2.3 states 0 is not the successor of any natural number. Hence, if $m, n$ are both positive, then $mn$ is positive.

Now consider the first statement. The left-to-right implication is simply the contrapositive of the second statement, which is true. Observe the right-to-left implication. If $n = 0$, then $0 \times m = 0$ by the definition of multiplication. If $m = 0$, then by commutativity of multiplication, $m \times 0 = 0 \times m = 0$. Hence, the statements are proven as desired. $\square$

**Exercise 2.3.3**. *Prove Proposition 2.3.5 (Associativity of multiplication): For any natural numbers $a, b, c$, we have $(a \times b) \times c = a \times (b \times c)$.*

*Solution.* We induct on $a$, keeping $b$ and $c$ fixed.

First, consider the base case $a = 0$. According to the definition of multiplication, the left-hand side leads to $(0 \times b) \times c = 0 \times c = 0$. Similarly, the right-hand side results in $0 \times (b \times c) = 0$. Because both sides are equal, our base case is proven.

Now we assume inductively that $(a \times b) \times c = a \times (b \times c)$; we need to show that $((a+1) \times b) \times c = (a+1)(b \times c)$. By the distributive law, the left-hand side results in $((a+1) \times b) \times c = (a \times b + b) \times c = (a \times b) \times c + b \times c$. The right-hand side produces $(a+1) \times (b \times c) = a \times (b \times c) + b \times c$, which, by the inductive hypothesis, is equal to $(a \times b) \times c + b \times c$. Thus, both sides are equal, and this closes the induction. $\square$

**Exercise 2.3.4**. *Prove the identity $(a+b)^2 = a^2 + 2ab + b^2$ for all natural numbers $a, b$.*

*Solution.* By Proposition 2.3.4 (distributive law) and Proposition 2.3.5 (commutativity of multiplication):

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b)\\ &= a(a+b)+b(a+b)\\ &= a^2+ab+ab+b^2\\ &= a^2+2ab+b^2\end{aligned}$$

Hence proven. □

**Exercise 2.3.5**. *Prove Proposition 2.3.9 (Euclidean algorithm): Let $n$ be a natural number, and let $q$ be a positive number. Then there exists natural numbers $m, r$ such that $0 \leq r < q$ and $n = mq + r$.*

*Solution.* We induct on $n$, fixing $q$.

First, observe the base case $n = 0$. There is an obvious solution of $m = 0, r = 0$. Hence, the base case is proven.

Now, suppose inductively that there exist natural numbers $m, r$ such that $0 \leq r < q$ and $n = mq + r$; we now have to show there exists natural numbers $m', r'$ such that $0 \leq r' < q$ and $n+1 = m'q+r'$. By the induction hypothesis, $n+1 = mq+r+1$. Since $r < q$, $r + 1 \leq q$. Thus, we have two cases:

(a) $r + 1 < q$: In this case, we can have $n + 1 = mq + (r + 1)$ with $0 \leq r + 1 < q$. Thus, we can have $m' = m$, $r' = r + 1$.

(b) $r + 1 = q$: In this case, we have $n + 1 = mq + q = (m + 1)q$ by the distributive law. Therefore, $m' = m + 1, r' = 0$.

This closes the induction. □

# 3 Set theory

## 3.1 Fundamentals

**Exercise 3.1.1**. *Show that the definition of equality in Definition 3.1.4 is reflexive, symmetric, and transitive.*

*Solution.* Let $A$, $B$, and $C$ be sets.

(a) *Reflexivity:* $A = A$

For every element $x$ of $A$, $x$ is clearly an element of $A$. Thus by Definition 3.1.4, $A = A$.

(b) *Symmetry: If* $A = B$, *then* $B = A$.

Suppose $A = B$. Then for every element $x \in B$, $x \in A$, and for every element $y \in A$, $y \in B$. Thus $B = A$.

(c) *Transitivity: If* $A = B$ *and* $B = C$, *then* $A = C$.

Suppose $A = B$ and $B = C$. Thus, for an arbitrary element $x \in A$, $x \in B$ because $A = B$, and since $B = C$, we see that $x \in C$. Similarly, for an arbitrary element $y \in C$, it follows that $y \in B$, and then $y \in A$. Hence, by Definition 3.1.4, it follows that $A = C$ if $A = B$ and $B = C$. $\square$

**Exercise 3.1.2**. *Using only Definition 3.1.4, Axiom 3.1, Axiom 3.2, and Axiom 3.3, prove that the sets* $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\}\}$, *and* $\{\varnothing, \{\varnothing\}\}$ *are all distinct (i.e., no two of them are equal to each other).*

*Solution.* First, let's show that $\varnothing$ is different from every other set. The empty set $\varnothing$ has no elements. However, $\varnothing$ is an element of $\{\varnothing\}$ and $\{\varnothing, \{\varnothing\}\}$, and $\{\varnothing\}$ is an element of $\{\{\varnothing\}\}$. Thus, $\varnothing$ is different from the other sets.

Now observe the set $\{\varnothing\}$. According to Axiom 3.3, $x \in \{\varnothing\}$ if and only if $x = \varnothing$. However, $\{\varnothing\}$ is an element of both $\{\{\varnothing\}\}$ and $\{\varnothing, \{\varnothing\}\}$. Hence, $\{\varnothing\}$ is different from the other sets.

We now need to show that $\{\{\varnothing\}\} \neq \{\varnothing, \{\varnothing\}\}$. Notice $\varnothing \notin \{\{\varnothing\}\}$ but $\varnothing \in \{\varnothing, \{\varnothing\}\}$. Therefore, $\{\{\varnothing\}\}$ and $\{\varnothing, \{\varnothing\}\}$ are distinct sets too. We conclude that the sets $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\}\}$, and $\{\varnothing, \{\varnothing\}\}$ are all distinct. $\square$

**Exercise 3.1.3**. *Prove the remaining claims in Lemma 3.1.13.*

*Solution.*

(a) *If $a$ and $b$ are objects, then $\{a, b\} = \{a\} \cup \{b\}$.*

By Axiom 3.4, for an object $x$, $x \in \{a\} \cup \{b\} \iff (x \in \{a\}$ or $x \in \{b\})$. Also, Axiom 3.3 states that $x \in \{a\} \iff x = a$. Similarly, $x \in \{b\} \iff x = b$. Axiom 3.3 also states that $x = a$ or $x = b$ if and only if $x \in \{a, b\}$. Therefore, if $x \in \{a\} \cup \{b\}$, $x \in \{a, b\}$. According to Definition 3.1.4, this implies $\{a, b\} = \{a\} \cup \{b\}$.

(b) *(Commutativity of the union operation) $A \cup B = B \cup A$.*

For any object $x$, $x \in A \cup B \iff (x \in A$ or $x \in B)$. If $x \in A$, then by Axiom 3.4, then $x \in B \cup A$. Similarly, if $x \in B$, then $x \in B \cup A$. We can show in a similar fashion that $x \in B \cup A \implies x \in A \cup B$. Hence, by Definition 3.1.4, $A \cup B = B \cup A$.

(c) $A \cup A = A \cup \varnothing = \varnothing \cup A = A$.

First, by Axiom 3.4, for an object $x$, $x \in A \cup A \iff (x \in A$ or $x \in A)$. But, $(x \in A$ or $x \in A)$ is equivalent to $x \in A$. So, $x \in A \cup A \iff x \in A$, thus $A \cup A = A$.

Now consider $A \cup \varnothing$. From Axiom 3.4, for an object $y$, $y \in A \cup \varnothing \iff (y \in A$ or $y \in \varnothing)$. However, Axiom 3.2 states that for an arbitrary $y$, $y \not in \varnothing$. Therefore, $y \in A \cup \varnothing \iff y \in A$, which implies $A \cup \varnothing = A$.

As we have proven the union operation is commutative, $A \cup \varnothing = \varnothing \cup A = A$. $\square$

**Exercise 3.1.4**. *Prove the remaining claims in Proposition 3.1.18.*

*Solution.*

(a) *If $A \subseteq B$ and $B \subseteq A$, then $A = B$.*

According to Definition 3.1.15, because $A \subseteq B$, for any object $x$, $x \in A \implies$

$x \in B$. Similarly, because $B \subseteq A$, $x \in B \implies x \in A$. Thus, $x \in A \iff x \in B$, which implies $A = B$.

(b) *If $A \subsetneq B$ and $B \subsetneq C$, then $A \subsetneq C$.*

Let $x$ be an arbitrary element of $A$. Since $A \subsetneq B$ and $B \subsetneq C$, it follows that $x \in B$, and thus $x \in C$. This shows that $A \subseteq C$. However, because $A \subsetneq B$, there exists an element $y \in B$ with $y \notin A$. Since $B \subsetneq C$, $y \in C$. Thus, we have an element $y$ where $y \in C$ and $y \notin A$. This shows that $A \neq C$. Hence, because $A \subseteq C$ and $A \neq C$, we can conclude that $A \subsetneq C$. $\square$

**Exercise 3.1.5**. *Let $A, B$ be sets. Show that the three statements $A \subseteq B$, $A \cup B = B$, $A \cap B = A$ are logically equivalent (any one of them implies the other two).*

*Solution.* First, we show that $A \subseteq B \implies A \cup B = B$. To prove $A \cup B = B$, we need to prove that if $A \subseteq B$, then $x \in A \cup B \iff x \in B$ for any object $x$. Clearly, $x \in B \implies x \in A \cup B$, so we only need to prove $x \in A \cup B \implies x \in B$. Because $A \subseteq B$, we know $x \in A \implies x \in B$. Thus, $x \in A \cup B$ implies $x \in A$ or $x \in B$, which is logically equivalent to $x \in B$. Therefore, $A \subseteq B \implies A \cup B = B$.

Now consider $A \cup B = B \implies A \cap B = A$. To prove $A \cap B = A$, we need to show that if $A \cup B = B$, then $x \in A \cap B \iff x \in A$. Proving $x \in A \cap B \implies x \in A$ is trivial based on Definition 3.1.23. Consider the converse implication $x \in A \implies x \in A \cap B$. Suppose $x \in A$. This implies $x \in A \cup B$, and because $A \cup B = B$, $x \in B$. By Definition 3.1.23, $x \in A$ and $x \in B \iff x \in A \cap B$. Therefore, $A \cup B = B \implies A \cap B = A$.

Lastly, we need to prove $A \cap B = A \implies A \subseteq B$. Because $A \cap B = A$, for all $x \in A, x \in A \cap B$. However, $x \in A \cap B \iff x \in A$ and $x \in B$. Therefore, $x \in A \implies x \in B$. $\square$

**Exercise 3.1.6**. *Prove Proposition 3.1.28 (Sets form a boolean algebra).*

*Solution.* Let $A, B, C$ be sets, and let $X$ be a set containing $A, B, C$ as subsets.

(a) *(Minimal element) We have $A \cup \varnothing = A$ and $A \cap \varnothing = \varnothing$.*

Consider $A \cup \varnothing$. From Axiom 3.4, for an object $y$, $y \in A \cup \varnothing \iff (y \in A$ or $y \in \varnothing)$. However, Axiom 3.2 states that for an arbitrary $y$, $y \notin \varnothing$. Therefore, $y \in A \cup \varnothing \iff y \in A$, which implies $A \cup \varnothing = A$.

Now consider $A \cap \varnothing$. For an object $y$, $y \in A \cap \varnothing \iff (y \in A \text{ or } y \in \varnothing)$. However, Axiom 3.2 states that for an arbitrary $y$, $y \notin \varnothing$. Therefore, no elements fulfil such conditions, or $A \cap \varnothing = \varnothing$.

(b) *(Maximal element) We have $A \cup X = X$ and $A \cap X = A$.*

Consider $A \cup X$. From Axiom 3.4, for an object $y$, $y \in A \cup X \iff y \in A$ or $y \in X$. However, if $y \in A$, then $y \in X$ because $A \subseteq X$. Thus, $y \in A \cup X \iff y \in A$, or $A \cup X = X$

Now consider $A \cap X = A$. From Definition 3.1.23, $y \in A \cap X \iff y \in A$ and $y \in X$. However, if $y \in A$, then $y \in X$ because $A \subseteq X$. Thus, $y \in A \cap X \iff y \in A$, or $A \cap X = A$.

(c) *(Identitiy) We have $A \cap A = A$ and $A \cup A = A$.*

First, consider $A \cap A = A$. Definition 3.1.23 states that if an arbitrary object $y \in A \cap A$, then $y \in A$ and $y \in A$. Thus, $y \in A \cap A \iff y \in A$, or $A \cap A = A$ as required. Similarly, for $A \cup A = A$, Axiom 3.4 states that if an arbitrary object $y \in A \cup A$, then $y \in A$ or $y \in A$. This leads to $y \in A \cup A \iff y \in A$, or $A \cup A = A$ as required.

(d) *(Commutativity) We have $A \cup B = B \cup A$ and $A \cap B = B \cap A$.*

Commutativity of the union operation, or $A \cup B = B \cup A$ has been proven in Exercise 3.1.3 (b). We will only show that the intersection operation is commutative, or $A \cap B = B \cap A$. Let $y$ be an object such that $y \in A \cap B$. Then $y \in A$ and $y \in B$. This is logically equivalent to $y \in B$ and $y \in A$. The converse is proven in a similar fashion, hence, $y \in A \cap B \iff y \in B \cup A$, or $A \cap B = B \cap A$.

(e) *(Associativity) We have $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.*

The proof of the associativity of the union operation can be found in Lemma 3.1.13. We will only prove the associativity of the intersection operation. By Definition 3.1.14, we need to prove that all elements of $(A \cap B) \cap C$ is an element of $A \cap (B \cap C)$. Suppose $y$ is an arbitrary element of $(A \cap B) \cap C$. Thus, by Definition 3.1.23, $y$ is both an element of $(A \cap B)$ and $C$. Because $y \in A \cap B$, $y \in A$ and $y \in B$. Since $y \in B$ and $y \in C$, then $y \in B \cap C$. Also, as $y \in A$, we can conclude that $y \in A \cap (B \cap C)$. The converse is proven similarly. Hence, $(A \cap B) \cap C = A \cap (B \cap C)$ as desired.

(f) *(Distributivity) We have* $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ *and* $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

First consider $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Suppose $x \in A \cap (B \cup C)$. Then, $x \in A$ and $x \in B \cup C$. So $x \in B$ or $x \in C$. We will now split into two cases. If $x \in B$, then $x \in A \cap B$, thus $x \in (A \cap B) \cup (A \cap C)$. Similarly, if $x \in C$, then $x \in A \cap C$, thus $x \in (A \cap B) \cup (A \cap C)$. Thus, $x \in A \cap (B \cup C) \implies x \in (A \cap B) \cup (A \cap C)$.

For the converse, suppose instead $x \in (A \cap B) \cup (A \cap C)$. Then, $x \in A \cap B$ or $x \in (A \cap C$. If $x \in A \cap B$, then $x \in A$ and $x \in B$. Because $x \in B$, we know $x \in B \cup C$. Therefore, $x \in A \cap (B \cup C)$. In a similar fashion, if $x \in A \cap C$, we can conclude that $x \in A \cap (B \cup C)$. Hence, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ as required.

We will now consider $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Suppose $x$ is an element of $A \cup (B \cap C)$. We know that $x \in A$ or $x \in (B \cap C)$. We will now split into two cases. If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$. If instead $x \in B \cap C$, then $x \in B$ and $x \in C$, so $x \in A \cup B$ and $x \in A \cup C$ as well. In both cases, we can conclude that $x \in (A \cup B) \cap (A \cup C)$.

(g) *(Partition) We have* $A \cup (X \setminus A) = X$ *and* $A \cap (X \setminus A) = \varnothing$.

We will first consider $A \cup (X \setminus A)$. We know that $x \in A$ or $x \in X \setminus A$. If $x \in A$, then $x \in X$ because $A \subseteq X$. If $x \in X \setminus A$ instead, then $x \in X$ and $x \notin A$. Either way, we obtain $x \in X$, thus $x \in A \cup (X \setminus A) \implies x \in X$.

Now consider the converse. Suppose instead $x \in X$. If $x \in A$, then $x \in A \cup (X \setminus A)$. If instead $x \notin A$, then $x \in X \setminus A$ because $x \in X$. Either way, $x \in X \implies x \in A \cup (X \setminus A)$. Hence, $A \cup (X \setminus A) = X$ as desired.

Consider instead $A \cap (X \setminus A)$. Suppose $x \in A \cap (X \setminus A)$. Then, $x \in A$ and $x \in X \setminus A$. Because $x \in X \setminus A$, $x \notin A$, which contradicts $x \in A$ obtained earlier. Thus, no element $x$ fulfills $x \in A \cap (X \setminus A)$, or $x \in \varnothing$. Hence $A \cap (X \setminus A) = \varnothing$.

(h) *(De Morgan Laws) We have* $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ *and* $X \setminus (A \cap B) =$

$(X \setminus A) \cup (X \setminus B)$.

First prove $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$. Suppose $x \in X \setminus (A \cup B)$. Then, $x \in X$ and $x \notin A \cup B$, or $x \notin A$ and $x \notin B$. Thus, we obtain $x \in X \setminus A$ and $x \in X \setminus B$. This leads to $x \in (X \setminus A) \cap (X \setminus B)$.

Now consider the converse. Suppose instead $x \in (X \setminus A) \cap (X \setminus B)$. Because $x \in X \setminus A$ and $X \setminus B$, $x \in X$, $x \notin A$ and $x \notin B$. Thus, $x \notin A \cup B$, and $x \in X \setminus (A \cup B)$. Hence, $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$.

We will now show that $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$. Suppose $x \in X \setminus (A \cap B)$. Then, $x \in X$ and $x \notin A \cap B$. Because $x \notin A \cap B$, $x \notin A$ or $x \notin B$. If $x \notin A$, then $x \notin A$ or $x \notin B$ since the former statement is true. Thus, $x \in (X \setminus A)$ or $x \in (X \setminus B)$, or $x \in (X \setminus A) \cup (X \setminus B)$.

Consider instead the converse. Suppose instead $x \in (X \setminus A) \cup (X \setminus B)$. Then, $x \in (X \setminus A)$ or $x \in (X \setminus B)$. If $x \in (X \setminus A)$, then $x \in X$ and $x \notin A$. Because $x \notin A$, $x \notin A \cap B$. Thus, $x \in X \setminus (A \cap B)$ because $x \in X$. Similarly, if $x \in (X \setminus B)$, $x \in X \setminus (A \cap B)$. Hence, we obtain $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$. $\square$

**Exercise 3.1.7**. *Let $A, B, C$ be sets. Show that $A \cap B \subseteq A$ and $A \cap B \subseteq B$. Furthermore, show that $C \subseteq A$ and $C \subseteq B$ if and only if $C \subseteq A \cap B$. In a similar spirit, show that $A \subseteq A \cup B$ and $B \subseteq A \cup B$, and furthermore that $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.*

*Solution.*

(a) *Show that $A \cap B \subseteq A$ and $A \cup B \subseteq B$.*

From Definition 3.1.23, for all objects $x \in A \cap B$, $x \in A$ and $x \in B$. Thus, $x \in A \cap B \implies x \in A$, which is equivalent to $A \cap B \subseteq A$. Similarly, using the commutativity of the union operation, $A \cup B \subseteq B$.

(b) *Show that $C \subseteq A$ and $C \subseteq B$ if and only if $C \subseteq A \cap B$.*

Suppose $x \in C$. We will first prove the implication. If $C \subseteq A$ and $C \subseteq B$, then $x \in A$ and $x \in B$. Therefore, $x \in A \cap B$, and $C \subseteq A \cap B$. Now consider the converse. If $C \subseteq A \cap B$, then $x \in A \cap B$. Thus, $x \in A$ and $x \in B$, or $C \subseteq A$ and $C \subseteq B$.

(c) *Show that $A \subseteq A \cup B$ and $B \subseteq A \cup B$.*

For all objects $x \in A$, $x \in A \cup B$, thus $A \subseteq A \cup B$. Similarly, for all objects $x \in B$, $x \in A \cup B$, thus $B \subseteq A \cup B$.

(d) *Show that $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.*

We will first prove the implication. Suppose $x \in A \cup B$. Then, $x \in A$ or $x \in B$. If $x \in A$, then $x \in C$ because $A \subseteq C$. Similarly, if $x \in B$, then $x \in C$. Therefore, $A \cup B \subseteq C$.

Now consider the converse. Suppose instead $x \in A$, thus $x \in A \cup B$. Because $A \cup B \subseteq C$, $x \in A \cup B \implies x \in C$. Thus $x \in A \implies x \in C$, or $A \subseteq C$. Similarly, by swapping the roles of $A$ and $B$ in the above argument, $B \subseteq C$. Hence, the statement is proven. □

**Exercise 3.1.8**. *Let $A, B$ be sets. Prove the* absorption laws *$A \cap (A \cup B) = A$ and $A \cup (A \cap B) = A$.*

*Solution.* First, consider $A \cap (A \cup B) = A$. Suppose $x$ is an element of $A \cap (A \cup B)$, thus $x \in A$ and $x \in A \cup B$. In particular, because $x \in A$, we obtain $A \cap (A \cup B) \subseteq A$. Instead, let $x \in A$, thus $x \in A \cup B$. This leads to $x \in A \cap (A \cup B)$, so $A \subseteq A \cap (A \cup B)$. Hence, $A \cap (A \cup B) = A$.

Now consider $A \cup (A \cap B) = A$. Using the distributivity of the union operation proven in Proposition 3.1.28 (f), the left-hand side is equivalent to $(A \cup A) \cap (A \cup B) = A \cap (A \cup B)$. From the previous results, this is equal to $A$. Therefore, we obtain $A \cup (A \cap B) = A$ as desired. □

**Exercise 3.1.9**. *Let $A, B, X$ be sets such that $A \cup B = X$ and $A \cap B = \varnothing$. Show that $A = X \setminus B$ and $B = X \setminus A$.*

*Solution.* Suppose $x \in A$. From $A \cup B = X$, we know $x \in A \cup B$, and thus $x \in X$. Also, from $A \cap B = \varnothing$, $A$ and $B$ are disjoint, therefore $x \notin B$. Thus, because $x \in X$ and $x \notin B$, $x \in X \setminus B$. Suppose instead that $x \in X \setminus B$. This implies $x \in X$ and $x \notin B$. Because $x \notin B$ and $x \in A \cup B$, $x \in A$. Hence, $A = X \setminus B$. In a similar fashion, we can swap the roles of $A$ and $B$ in the above proof to show $B = X \setminus A$. □

**Exercise 3.1.10**. *Let $A$ and $B$ be sets. Show that the three sets $A \setminus B$, $A \cap B$ and $B \setminus A$ are disjoint, and that their union is $A \cup B$.*

*Solution.* To show the three sets are mutually disjoint, we prove that the intersections of any two of the sets have no elements. Elements in $A \setminus B$ are in $A$ and not $B$. However, elements in $A \cap B$ are in both $A$ and $B$, and elements in $B \setminus A$ are in set $B$ and not $A$. Because an object cannot both be in a set and not in a set, all three intersections are empty sets.

We will now prove that the union of the three sets is $A \cup B$. Consider the following cases for an object $x$:

(a) $x \in A$, $x \in B$: $x \in A \cap B$;

(b) $x \in A$, $x \notin B$: $x \in A \setminus B$;

(c) $x \notin A, x \in B$: $x \in B \setminus A$,

(d) $x \notin A, x \notin B$: $x$ is not an element of any of the sets, and is therefore also not in the union.

Thus, if $x \in A$ or $x \in B$, $x$ is an element of exactly one of the sets. This is logically equivalent to saying $A \cup B$ is the union of the three sets as per Axiom 3.4. $\square$

**Exercise 3.1.11**. *Show that the axiom of replacement implies the axiom of specification.*

*Solution.* Let $A$ be a set, and $P(x)$ be a property pertaining to $x$. To prove the axiom of specification, we need to construct the set $\{x \in A : P(x)\}$ using only the axiom of replacement. Let $Q(x, y)$ be the property that $y = x$ and $P(x)$ is true. To use the axiom of replacement, for each $x \in A$, there must be at most one $y$ such that $Q(x, y)$ is true. Indeed, for each $x \in A$, there is only one $y$ such that $y = x$. Therefore, there is at most one $y$ such that $Q(x, y)$ is true. We can now show that there exists a set $\{y : Q(x, y) \text{ for some } x \in A\}$ such that for an object $z$,

$$
\begin{aligned}
& z \in \{y : Q(x, y) \text{ for some } x \in A\} \\
\iff & Q(x, z) \text{ is true for some } x \in A \\
\iff & x = z \text{ and } P(x) \text{ is true for some } x \in A \\
\iff & P(x) \text{ for some } x \in A \\
\iff & P(z) \text{ for some } z \in A.
\end{aligned}
$$

Thus, we can construct the set $\{x \in A : P(x)\}$, which means we have proven the axiom of specification. $\square$

## 3.2 Russell's Paradox

**Exercise 3.2.1**. *Show that the universal specification axiom, Axiom 3.8, if assumed to be true, would imply Axioms 3.2, 3.3, 3.4, 3.5, and 3.6. (If we assume that all natural numbers are objects, we also obtain Axiom 3.7.)*

*Solution.* Recall that Axiom 3.8 states that if for every object $x$ we have a property $P(x)$ pertaining to $x$, then there exists a set $\{x : P(x)\}$ such that for every object $y$, $y \in \{x : P(x)\} \iff P(y)$.

(a) *Axiom 3.2: Empty set.*

Let $P(x)$ be a property pertaining to an object $x$ that is always false. From Axiom 3.8, the set $\{x : P(x)\}$ exists. However, for any object $y$, $P(y)$ is false. It follows that $y \in \{x : P(x)\}$ is false for all objects $y$. Therefore, the set $\{x : P(x)\}$ has no elements. This is precisely the claim of Axiom 3.2.

(b) *Axiom 3.3: Singleton sets and pair sets.*

Let $a$ be an arbitrary object and $P(x)$ be the property pertaining to $x$ such that $x = a$. From Axiom 3.8, the set $\{x : x = a\}$ exists, and it has exactly one element $a$. Thus, the singleton set $\{a\}$ exists.

Suppose instead $a$ and $b$ are arbitrary objects, and let $Q(x)$ be the property pertaining to $x$ such that $x = a$ or $x = b$. From Axiom 3.8, the set $\{x : x = a \text{ or } x = b\}$ exists, which has exactly two elements $a$ and $b$. Thus, the pair set $\{a, b\}$ exists.

(c) *Axiom 3.4: Pairwise union.*

Let $A$ and $B$ be sets, and $x$ be an object. Write $P(x)$ as the property pertaining to $x$ that $x \in A$ or $x \in B$. Thus, there exists a set $\{x : x \in A \text{ or } x \in B\}$. We denote this set as the union of sets $A$ and $B$, or $A \cup B$. For any object $y$, $(y \in A \text{ or } y \in B) \iff y \in A \cup B$. This is precisely the claim of Axiom 3.4.

(d) *Axiom 3.5: Axiom of specification.*

Let $A$ be a set, and let $P(x)$ be a property pertaining to an object $x \in A$. Let $Q(x)$ be the property that $P(x)$ is true and $x \in A$. By Axiom 3.8, the set $\{x : Q(x)\}$ exists. For any object $y$, $y \in \{x : Q(x)\} \iff (y \in A \text{ and } P(y))$. This is exactly the claim of Axiom 3.5.

(e) *Axiom 3.6: Replacement.*

Let $A$ be a set, and $P(x, y)$ be a property pertaining to $x$ and $y$. Let $Q(y)$ be a property pertaining to $y$ that $P(x, y)$ is true for some $x \in A$. According to Axiom 3.8, there exists a set $\{y : P(x, y) \text{ for some } x \in A\}$, such that for any object $z$, $z \in \{y : P(x, y) \text{ for some } x \in A\} \iff P(x, z)$ for some $x \in A$. This is exactly the claim of Axiom 3.6.

(f) *Axiom 3.7: Infinity.*

Suppose all natural numbers are objects. Let $P(x)$ be the statement that $x$ is a natural number. Then the set $\{x : P(x)\}$ exists. We denote this set as $\mathbb{N}$. □

**Exercise 3.2.2**. *Use the axiom of regularity (and the singleton set axiom) to show that if $A$ is a set, then $A \notin A$. Furthermore, show that if $A$ and $B$ are two sets, then either $A \notin B$ or $B \notin A$ (or both).*

*Solution.* We will first prove that if $A$ is a set, then $A \notin A$. Consider the set $\{A\}$ which exists by the singleton set axiom. By the axiom of regularity, there is at least one element $x \in \{A\}$ such that $x$ is either not a set, or is disjoint from $\{A\}$. From the singleton set axiom, we know that $x = A$. However, $A$ is a set, so $A$ and $\{A\}$ must be disjoint sets, i.e. $A \cap \{A\} = \varnothing$. If $A \in A$, then $A$ is an element of $A \cap \{A\}$, which is contradictory. Therefore $A \notin A$ as desired.

We will now show that if $A$ and $B$ are two sets, then either $A \notin B$ or $B \notin A$ (or both). Consider the set $\{A, B\}$, which exists by the pair set axiom. According to the axiom of regularity, there is at least one element $x \in \{A, B\}$ such that $x$ is either not a set, or is disjoint from $\{A, B\}$. Since both $A$ and $B$ are sets, $x$ must be a set, thus $x$ must be disjoint from $\{A, B\}$. If $x = A$, then $A$ is disjoint from $\{A, B\}$. Thus, for any object $y$, $y \notin A \cap \{A, B\}$, in particular, $B \notin A \cap \{A, B\}$. Since $B \in \{A, B\}$, it follows that $B \notin A$. Similarly, if $x = B$, we can deduce that $A \notin B$. Hence, either $B \notin A$ or $A \notin B$ as required. □

**Exercise 3.2.3**. *Show (assuming the other axioms of set theory) that the universal specification axiom, Axiom 3.8, is equivalent to an axiom postulating the existence of a "universal set" $\Omega$ consisting of all objects (i.e., for all objects $x$, we have $x \in \Omega$). In other words, if Axiom 3.8 is true, then a universal set exists, and conversely, if a universal set exists, then Axiom 3.8 is true.*

*Solution.* We will first show the statement. Suppose Axiom 3.8 is true. Then, let $P(x)$ be a property pertaining to $x$ such that $x = x$. Thus, there exists a set $\{x : x = x\}$ such that for every object $y$, $y \in \{x : x = x\} \iff P(y)$. Because $P(y)$ is true for all $y$, all objects are elements of $\{x : x = x\}$. Thus, the set $\{x : x = x\}$ is a universal set.

Now consider the converse. Suppose all objects are in a universal set $\Omega$. We want to show that there exists a set $\{x : P(x)\}$ such that for every object $y$, $y \in \{x : P(x)\} \iff P(y)$. Let $A$ be a set, and $P(x)$ be the property pertaining to $x$. By the axiom of specification, there exists a set $\{x \in A : P(x)\}$ such that $y \in \{x \in A : P(x)\} \iff (y \in A \text{ and } P(y))$. Take $A = \Omega$. It follows that $x \in A$ for all $x$. Therefore, there exists a set $\{x : P(x)\}$ such that $y \in \{x : P(x)\} \iff P(y)$, which is exactly the claim of Axiom 3.8. □

## 3.3 Functions

**Exercise 3.3.1**. *Show that the definition of equality in Definition 3.3.7 is reflexive, symmetric, and transitive. Also verify the substitution property: if $f, \tilde{f} : X \to Y$ and $g, \tilde{g} : Y \to Z$ are functions such that $f = \tilde{f}$ and $g = \tilde{g}$, then $g \circ f = \tilde{g} \circ \tilde{f}$.*

*Solution.* For the first section of this exercise, let $X$ and $Y$ be sets, and $f : X \to Y$, $g : X \to Y$, and $h : X \to Y$ be functions of the same domain and range.

(a) *Reflexivity: $f = f$.*

Notice that $f$ and $f$ have the same domain and range. For every value $x \in X$, clearly $f(x) = f(x)$ by reflexivity of elements in set $Y$. Thus, by Definition 3.3.7, $f = f$.

(b) *Symmetry: If $f = g$, $g = f$.*

Suppose $f = g$. Then $f(x) = g(x)$ forall $x \in X$. But, from the symmetry of elements in set $Y$, for all $x \in X$, $g(x) = f(x)$. Heence $g = f$.

(c) *Transitivity: If $f = g$ and $g = h$, then $f = h$.*

Suppose $f = g$ and $g = h$. Then, $f(x) = g(x)$ and $g(x) = h(x)$. For any $x \in X$, $f(x) = g(x)$ and $g(x) = h(x)$. By the transitivity of elements in set $Y$, we obtain $f(x) = h(x)$. Since $x$ is arbitrary, we can conclude that $f = h$.

Suppose instead $X, Y, Z$ are sets, and suppose also $f, \tilde{f} : X \to Y$ and $g, \tilde{g} : Y \to Z$ such that $f = \tilde{f}$ and $g = \tilde{g}$. Thus, by Definition 3.3.7, $f(x) = \tilde{f}(x)$ for all $x \in X$ and $g(y) = \tilde{g}(y)$ for all $y \in Y$. By the substitution axiom, $g(f(x)) = g(\tilde{f}(x))$. Since $g = \tilde{g}$, $g(f(x)) = g(\tilde{f}(x)) = \tilde{g}(\tilde{f}(x))$, or $(g \circ f)(x) = (\tilde{g} \circ \tilde{f})(x)$. By Definition 3.3.7, we obtain $g \circ f = \tilde{g} \circ \tilde{f}$ as required. □

**Exercise 3.3.2**. *Let $f : X \to Y$ and $g : Y \to Z$ be functions. Show that if $f$ and $g$ are both injective, then so is $g \circ f$; similarly, show that if $f$ and $g$ are both surjective, then so is $g \circ f$.*

*Solution.* Note that $g \circ f : X \to Z$.

First, we will show that if $f$ and $g$ are both injective, then so is $g \circ f$. Suppose $g(f(x)) = g(f(x'))$, where $x, x' \in X$. Then, by injectivity of $g$, we obtain $f(x) = f(x')$. Similarly, by injectivity of $f$, we obtain $x = x'$. Therefore, $g \circ f$ is injective.

We will now prove if $f$ and $g$ are both surjective, then so is $g \circ f$. Because $f$ is surjective, for every $y \in Y$, there exists an $x \in X$ such that $f(x) = y$. Similarly, because $g$ is surjective, for every $z \in Z$, there exists an $y \in Y$ such that $g(y) = z$. Combining these two results, for every $z \in Z$, there exists an $x \in X$ such that $g(f(x)) = z$. Therefore, by definition, $g \circ f$ is surjective. □

**Exercise 3.3.3**. *When is the empty function injective? Surjective? Bijective?*

*Solution.* Let $X$ be a set, and $f : \varnothing \to X$ be the empty function.

The function $f$ is considered **injective** if for any $p, p' \in \varnothing$ where $p \neq p'$, then $f(p) \neq f(p')$. This statement is vacuously true because there is no such $p$ and $p'$ in the empty set $\varnothing$. Thus, $f$ is always injective.

The function $f$ is considered **surjective** if for any value $x$ in $X$, there is a $p \in \varnothing$ such that $f(p) = x$. This statement is false if $X \neq \varnothing$ as the empty set $\varnothing$ has no elements, there is no such $p$. However, when $X = \varnothing$, this statement is vacuously true because no such $x$ exists. Thus, $f$ is surjective only when $X = \varnothing$.

Since $f$ is always injective and surjective only when $X = \varnothing$, $f$ is only **bijective** when $X = \varnothing$. $\square$

**Exercise 3.3.4**. *In this section we give some cancellation laws for composition. Let $f : X \to Y$, $\tilde{f} : X \to Y$, $g : Y \to Z$, and $\tilde{g} : Y \to Z$ be functions. Show that if $g \circ f = g \circ \tilde{f}$ and $g$ is injective, then $f = \tilde{f}$. Is the same statement true if $g$ is not injective? Show that if $g \circ f = \tilde{g} \circ f$ and $f$ is surjective, then $g = \tilde{g}$. Is the same statement true if $g$ is not surjective?*

*Solution.* We will first solve the first part of the exercise. Because $g$ is injective, if $g(f(x)) = g(\tilde{f}(x))$, then $f(x) = \tilde{f}(x)$ for any $x \in X$. Because $f$ and $\tilde{f}$ has the same domain and range, by Definition 3.3.7 (equality of functions), $f = \tilde{f}$. This statement is not true if $g$ is not injective. As a counterexample, take $g$ as a constant function, that is, $g(y) = c$ where $y \in Y$. Then, $g(f(x)) = g(\tilde{f}(x))$ for any $f$ and $\tilde{f}$, including when $f \neq \tilde{f}$.

Now, consider the second part of the exercise. Because $f$ is surjective, for any $y \in Y$, there exists $x \in X$ such that $y = f(x)$. Then, $g(y) = \tilde{g}(y)$ for any $y \in Y$. Because $g$ and $\tilde{g}$ has the same domain and range, $g = \tilde{g}$. This statement is not true if $f$ is not surjective. As a counterexample, take $f$ as a constant function, that is, $f(x) = c$, where $x \in X$, $c \in Y$, and $Y \neq \{c\}$. Evidently $f$ is not surjective. Thus, $g(c) = \tilde{g}(c)$ implies $g$ and $\tilde{g}$ are equal at $c$, but not necessarily at any other points. For instance, let $f(x) = 0$, $g(x) = x$, $\tilde{g}(x) = 2x$. Then, $g \neq \tilde{g}$ even though $g(0) = \tilde{g}(0)$. $\square$

**Exercise 3.3.5**. *Let $f : X \to Y$ and $g : Y \to Z$ be functions. Show that if $g \circ f$ is injective, then $f$ must be injective. Is it true that $g$ must also be injective? Show that if $g \circ f$ is surjective, then $g$ must be surjective. Is it true that $f$ must also be surjective?*

*Solution.* We will first solve the first part of the exercise. For the sake of contradiction, suppose $f$ isn't injective. In other words, suppose there exists objects $x, x' \in X$ such that $x \neq x'$ and $f(x) = f(x')$. This leads to $g(f(x)) = g(f(x'))$. But $x \neq x'$, which implies $g \circ f$ is not injective, which is contradictory. Thus, $f$ **must be injective**. However, $g$ need not be injective. As a counterexample, let $X = \{0, 1\}$, $Y = Z = \{0, 1, 2\}$ and $f(0) = 0$, $f(1) = 1$. Let $g$ such that $g(0) = 0, g(1) = 1, g(2) = 0$ such that $g \circ f$ is injective. Evidently, $g$ is not injective.

Now consider the second part of the exercise. For the sake of contradiction, suppose $g$ is not surjective. In other words, there exists a $z \in Z$ such that for all

$y \in Y$, $g(y) \neq z$. In particular, since $f(x) \in Y$, for all $x \in X$, $g(f(x)) \neq z$. This contradicts the condition that $g \circ f$ is surjective. Hence, $g$ **must be surjective**. However, $f$ need not be surjective. As a counterexample, let $X = Y = \{0, 1\}$, $Z = \{0\}$. We define $f$ such that $f(0) = 0$, $f(1) = 0$, and $g$ such that $g(0) = 0$, $g(1) = 0$. In this case, $g \circ f$ is surjective, but $f$ is not. □

**Exercise 3.3.6**. *Let $f : X \to Y$ be a bijective function, and let $f^{-1} : Y \to X$ be its inverse. Verify the cancellation laws $f^{-1}(f(x)) = x$ for all $x \in X$ and $f(f^{-1}(y)) = y$ for all $y \in Y$. Conclude that $f^{-1}$ is also invertible, and has $f$ has its inverse (thus $(f^{-1})^{-1} = f$).*

*Solution.* Suppose $x$ is an arbitrary element of $X$, and $y = f(x)$, $y \in Y$. By the definition of inverse functions, because $f$ is bijective, for every $y \in Y$, there is exactly one $x' \in X$ such that $y = f(x')$, which is $x' = f^{-1}(y)$. Because $f(x) = y = f(x')$, the injectivity of $f$ leads to $x = x'$. Thus, $x = f^{-1}(y) = f^{-1}(f(x))$ as required.

Suppose instead $y$ be an arbitrary element of $Y$. Since $f$ is bijective, there is exactly one value of $x \in X$ such that $f(x) = y$. This value of $x$ is denoted as $f^{-1}(y)$. Thus, $f(f^{-1}(y)) = y$ as desired.

Now, to show that $f^{-1}$ is also invertible, we need to show that $f^{-1}$ is bijective. In other words, we need to prove the injectivity and surjectivity of $f^{-1}$. Suppose $f^{-1}(y) = f^{-1}(y')$ for $y, y' \in Y$. Applying $f$ to both sides, we obtain $f(f^{-1}(y)) = f(f^{-1}(y'))$, which leads to $y = y'$ from previous results. Thus, $f^{-1}$ is injective. Because $f$ is bijective, for every $x \in X$, there exists $y \in Y$ such that $f(x) = y$. This is precisely the definition for $f^{-1}$ to be surjective. Hence, we have shown that $f^{-1}$ is bijective as required. □

**Exercise 3.3.7**. *Let $f : X \to Y$ and $g : Y \to Z$ be functions. Show that if $f$ and $g$ are bijective, then so is $g \to f$, and we have $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.*

*Solution.* From Exercise 3.3.2, we know $g \circ f$ is both injective and surjective, thus $g \circ f$ is bijective. In this exercise, we only need to show $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Let $z$ be an arbitrary element of $Z$. Because $g$ is bijective, for every $z \in Z$, there exists exactly one $y \in Y$ such that $g(y) = z$, or $y = g^{-1}(z)$. Similarly, because $f$ is bijective, for every $y \in Y$, there exists exactly one $x \in X$ such that $f(x) = y$, or $x = f^{-1}(y)$. Thus, for every $z \in Z$, there exists exactly one $x \in X$ such that $(g \circ f)(x) = z$, $x = f^{-1}(g^{-1}(z))$. Therefore, $x = (g \circ f)^{-1}(z) = (f^{-1} \circ g^{-1})(z)$, or $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ as desired. □

**Exercise 3.3.8**. *If $X$ is a subset of $Y$, let $\iota_{X\to Y} : X \to Y$ be the* inclusion map from $X$ to $Y$ *defined by mapping $x \mapsto x$for all $x \in X$, i.e. $\iota_{X\to Y}(x) := x$ for all $x \in X$. The map $\iota_{X\to X}$ in particular is called the* identity map *on $X$.*

*Solution.*

(a) *Show that if $X \subseteq Y \subseteq Z$ then $\iota_{Y\to Z} \circ \iota_{X\to Y} = \iota_{X\to Z}$.*

Note that $\iota_{Y\to Z}\circ\iota_{X\to Y}$ and $\iota_{X\to Z}$ have the same domain and range, and that the range of $\iota_{X\to Y}$ is equal to the domain of $\iota_{Y\to Z}$. Let $x$ be an arbitrary element of $X$. Thus, $\iota_{Y\to Z}(\iota_{X\to Y}(x)) = \iota_{Y\to Z}(x) = x$ because $x \in Y$. Also, $\iota_{X\to Z}(x) = x$. Hence, $\iota_{Y\to Z}(\iota_{X\to Y}(x)) = \iota_{X\to Z}(x)$ for all $x \in X$, thus $\iota_{Y\to Z} \circ \iota_{X\to Y} = \iota_{X\to Z}$.

(b) *Show that if $f : A \to B$ is any function, then $f = f \circ \iota_{A\to A} = \iota_{B\to B} \circ f$.*

The compositions $f\circ\iota_{A\to A}$ and $\iota_{B\to B}\circ f$ have the same domain and range as $f$, so we just have to prove these functions agree on all inputs. Let $a$ be an arbitrary element of $A$. Then, $f(\iota_{A\to A}(a)) = f(a)$. Also, $\iota_{B\to B}(f(a)) = f(a)$. As all three functions have the same output for all $a \in A$, $f = f \circ \iota_{A\to A} = \iota_{B\to B} \circ f$.

(c) *Show that, if $f : A \to B$ is a bijective function, then $f \circ f^{-1} = \iota_{B\to B}$ and $f^{-1} \circ f = \iota_{A\to A}$.*

Let $b$ be an arbitrary element of $B$. Since $f$ is bijective, there exists exactly one $a \in A$ such that $f(a) = b$. Thus, $f(f^{-1}(b)) = f(a) = b = \iota_{B\to B}(b)$ for all $b \in B$. Because $f \circ f^{-1}$ and $\iota_{B\to B}$ have the same domain and range, $f \circ f^{-1} = \iota_{B\to B}$. In a similar fashion, we can show that $f^{-1} \circ f = \iota_{A\to A}$ as required.

(d) *Show that if $X$ and $Y$ are disjoint sets, and $f : X \to Z$ and $g : Y \to Z$ are functions, then there is a unique function $h : X\cup Y \to Z$ such that $h\circ\iota_{X\to X\cup Y} = f$ and $h \circ \iota_{Y\to X\cup Y} = g$.*

We will first show the existence of the function $h$. Define $h$ as such: for any $p \in X \cup Y$, if $p \in X$, then $h(p) := f(p)$; if $p \in Y$, then $h(p) := g(p)$. Because $p$ must be in exactly one of $X$ and $Y$, this definition makes sense. Thus, $h(\iota_{X\to X\cup Y}(x)) = h(x) = f(x)$ for any $x \in X$ and $h(\iota_{Y\to X\cup Y}(y)) = g(y)$ for any $y \in Y$. This matches the given definition of $h$, hence the function $h$ exists.

We will now show the uniqueness of the function $h$. Suppose $\tilde{h} : X \cup Y \to Z$ is another function such that $\tilde{h} \circ \iota_{X\to X\cup Y} = f$ and $\tilde{h} \circ \iota_{Y\to X\cup Y} = g$. Let $p$ be an element in $X \cup Y$. If $p \in X$, then $\iota_{X\to X\cup Y}(p) = p$, thus $\tilde{h}(p) = \tilde{h}(\iota_{X\to X\cup Y}(p)) = f(p)$. However, this is equal to $h(p)$. Similarly, if $p \in Y$, $\tilde{h}(p) = h(p)$. Because $\tilde{h}(p) = h(p)$ for all $p \in X \cup Y$, $\tilde{h} = h$. Hence, the function $h$ is unique. □

## 3.4 Images and inverse images

**Exercise 3.4.1**. *Let $f : X \to Y$ be a bijective function, and let $f^{-1} : Y \to X$ be its inverse. Let $V$ be any subset of $Y$. Prove the forward image of $V$ under $f^{-1}$ be the same set as the inverse image of $V$ under $f$; thus the fact that both sets are denoted by $f^{-1}(V)$ will not lead to any inconsistencies.*

*Solution.* Let $F$ be the forward image of $V$ under $f^{-1}$, and $I$ be the inverse image of $V$ under $f$. Thus, by Definition 3.4.1 and Definition 3.4.4, $F = \{f^{-1}(x) : x \in V\}$ and $I = \{x \in X : f(x) \in V\}$ respectively. We need to show $F = I$, or $F \subseteq I$ and $I \subseteq F$.

We will first show $F \subseteq I$. Let $a$ be an arbitrary element of $F$. Then, there exists a $b \in V$ such that $a = f^{-1}(b)$, or $b = f(a)$, thus $f(a) \in V$. Also, because $F \subseteq X$ and $a = f^{-1}(b)$, $a \in X$. Thus, we have both $a \in X$ and $f(a) \in V$, which implies $a \in I$, hence $F \subseteq I$.

Consider instead the statement $I \subseteq F$. Suppose $c \in I$, then $f(c) \in V$. Write $d = f(c)$. Thus, we have $c \in I$ and $c = f^{-1}(d)$ for some $d \in V$. This implies $c \in F$ by definition of forward images, hence $I \subseteq F$.

**Exercise 3.4.2**. *Let $f : X \to Y$ be a function from one set $X$ to another set $Y$, let $S$ be a subset of $X$, and let $U$ be a subset of $Y$. What, in general, can one say about $f^{-1}(f(S))$ and $S$? What about $f(f^{-1}(U))$ and $U$?*

*Solution.* We will consider the containment of each pair of sets. Generally, we claim $S \neq f^{-1}(f(S))$, but $S \subseteq f^{-1}(f(S))$; and $f(f^{-1}(U)) \neq U$, but $f(f^{-1}(U)) \subseteq U$.

First consider $f^{-1}(f(S))$ and $S$. To show $f^{-1}(f(S)) \subseteq S$ is generally false, we will consider a counterexample. Let $f : \{0, 1\} \to \{1\}$ such that $f(0) = f(1) = 1$, and $S = \{0\}$. Thus, $f(S) = \{1\}$ and $f^{-1}(f(S)) = \{0, 1\}$, which is not a subset of $S$. Now, to prove $S \subseteq f^{-1}(f(S))$, suppose $x \in S$, thus $x \in X$. Then, by Definition 3.4.1, $f(x) \in f(S)$. This leads to $x \in f^{-1}(f(S))$ by Definition 3.4.4 because $x \in X$ and $f(x) \in f(S)$, which implies $S \subseteq f^{-1}(f(S))$ as desired.

Now consider $f(f^{-1}(U))$ and $U$. To show $U \subseteq f(f^{-1}(U))$ is generally false, we will consider a counterexample. Let $f : \{0\} \to \{0,1\}$ such that $f(0) = 1$, and $U = \{0,1\}$. Thus, $f^{-1}(U) = \{0\}$ and $f(f^{-1}(U)) = \{1\}$, which $U$ is not a subset of. However, to show $f(f^{-1}(U)) \subseteq U$, suppose $x \in f(f^{-1}(U))$. Then, there exists some $y \in f^{-1}(U)$ such that $x = f(y)$. According to Definition 3.4.4, $y \in f^{-1}(U)$ if and only if $f(y) \in U$, thus $x \in U$ because $x = f(y)$. Because $x \in U$ for all $x \in f(f^{-1}(U))$, we obtain $f(f^{-1}(U)) \subseteq U$ as desired. $\square$

**Exercise 3.4.3.** *Let $A, B$ be two subsets of a set $X$, and let $f : X \to Y$ be a function. Show that $f(A \cap B) \subseteq f(A) \cap f(B)$, that $f(A) \setminus f(B) \subseteq f(A \setminus B)$, $f(A \cup B) = f(A) \cup f(B)$. For the first two statements, is it true that the $\subseteq$ relation can be improved to $=$?*

*Solution.*

(a) $f(A \cap B) \subseteq f(A) \cap f(B)$.

Let $y \in f(A \cap B)$. Then, there exists some $x \in A \cap B$ such that $y = f(x)$. This implies $x \in A$ and $x \in B$. Because $y = f(x)$, $y \in f(A)$ and $y \in f(B)$, which leads to $y \in f(A) \cap f(B)$. Because $y \in f(A) \cap f(B)$ for all $y \in f(A \cap B)$, $f(A \cap B) \subseteq f(A) \cap f(B)$.

Consider the reverse inclusion instead. As a counterexample, let $A = \{0\}$, $B = \{1\}$, $f(x) = 5$. Thus, $A \cap B = \varnothing$, and $f(\varnothing) = \varnothing$. However, $f(A) = \{5\}$, $f(B) = \{5\}$, and $f(A) \cap f(B) = \{5\}$, which is not a subset of $f(A \cap B) = \varnothing$. Hence, the reverse inclusion is false.

(b) $f(A) \setminus f(B) \subseteq f(A \setminus B)$.

Let $y \in f(A) \setminus f(B)$; thus $y \in f(A)$ and $y \notin f(B)$. Then, there exists an $x \in A$ such that $y = f(x)$. If $x \in B$, then $y \in f(B)$, therefore $x \notin B$. Thus, we have $y = f(x)$ for some $x \in A \setminus B$, or $y \in f(A \setminus B)$ for all $y \in f(A) \setminus f(B)$. Hence, $f(A) \setminus f(B) \subseteq f(A \setminus B)$.

Consider the reverse inclusion instead. As a counterexample, let $A = \{1, 2\}$, $B = \{1\}$, and $f(x) = 4$ for any $x$. Thus, $f(A) = f(B) = \{4\}$, and $f(A) \setminus f(B) = \varnothing$. However, $A \setminus B = \{2\}$, and $f(A \setminus B) = \{4\}$, which is not a subset of $f(A) \setminus f(B) = \varnothing$. Hence, the reverse inclusion is false.

(c) $f(A \cup B) = f(A) \cup f(B)$.

We will first consider $f(A \cup B) \subseteq f(A) \cup f(B)$. Let $y \in f(A \cup B)$. Then, there exists an $x \in A \cup B$ such that $y = f(x)$. Thus, $x \in A$ or $x \in B$. If $x \in A$, then $y = f(x) \in f(A)$, which implies $y \in f(A) \cup f(B)$. Similarly, if $x \in B$, $y \in f(A) \cup f(B)$. Because $y \in f(A) \cup f(B)$ for all $y \in f(A \cup B)$, we can conclude that $f(A \cup B) \subseteq f(A) \cup f(B)$.

Now consider the reverse inclusion. Let $y \in f(A) \cup f(B)$. Then, $y \in f(A)$ or $y \in f(B)$. If $y \in f(A)$, then there exists an $x \in A$ such that $y = f(x)$. Therefore, $x \in A \cup B$, and it follows that $y \in f(A \cup B)$. Similarly, if $y \in f(B)$, then $y \in f(A \cup B)$. Because $y \in f(A \cup B)$ for all $y \in f(A) \cup f(B)$, we can conclude that $f(A) \cup f(B) \subseteq f(A \cup B)$. Hence, $f(A \cup B) = f(A) \cup f(B)$. $\square$

**Exercise 3.4.4**. *Let $f : X \to Y$ be a function from one set $X$ to another set $Y$, and let $U$, $V$ be subsets of $Y$. Show that $f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V)$, that $f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$, and that $f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$.*

*Solution.*

(a) $f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V)$.

Suppose $x \in f^{-1}(U \cup V)$. By definition, $f(x) \in U \cup V$; it follows that $f(x) \in U$ or $f(x) \in V$. If $f(x) \in U$, then $x \in f^{-1}(U)$, and $x \in f^{-1}(U) \cup f^{-1}(V)$. Similarly, if $f(x) \in V$, then $x \in f^{-1}(U) \cup f^{-1}(V)$. Therefore, for any $x \in f^{-1}(U \cup V)$, we have $x \in f^{-1}(U) \cup f^{-1}(V)$.

Now consider the converse. Suppose $x \in f^{-1}(U) \cup f^{-1}(V)$, thus $x \in f^{-1}(U)$ or $x \in f^{-1}(V)$. If $x \in f^{-1}(U)$, then $f(x) \in U$, and $f(x) \in U \cup V$. Similarly, if $x \in f^{-1}(V)$, then $f(x) \in U \cup V$. This leads to $x \in f^{-1}(U \cup V)$ for all $x \in f^{-1}(U) \cup f^{-1}(V)$. Hence, $f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V)$.

(b) $f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$.

Suppose $x \in f^{-1}(U \cap V)$. By definition, $f(x) \in U \cap V$; it follows that $f(x) \in U$ and $f(x) \in V$. Therefore, $x \in f^{-1}(U)$ and $x \in f^{-1}(V)$, thus $x \in f^{-1}(U) \cap f^{-1}(V)$.

Now consider the converse. Suppose $x \in f^{-1}(U) \cap f^{-1}(V)$, thus $x \in f^{-1}(U)$ and $x \in f^{-1}(V)$. By definition, $f(x) \in U$ and $f(X) \in V$, thus $f(x) \in U \cap V$.

This leads to $x \in f^{-1}(U \cap V)$ for all $x \in f^{-1}(U) \cap f^{-1}(V)$. Hence, $f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$.

(c) $f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$.

Suppose $x \in f^{-1}(U \setminus V)$. By definition, $f(x) \in U \setminus V$; it follows that $f(x) \in U$ and $f(x) \notin V$. Therefore, $x \in f^{-1}(U)$ and $x \notin f^{-1}(V)$, thus $x \in f^{-1}(U) \setminus f^{-1}(V)$.

Now consider the converse. Suppose $x \in f^{-1}(U) \setminus f^{-1}(V)$, thus $x \in f^{-1}(U)$ and $x \notin f^{-1}(V)$. By definition, $f(x) \in U$ and $f(x) \notin V$, thus $f(x) \in U \setminus V$. This leads to $x \in f^{-1}(U \setminus V)$ for all $x \in f^{-1}(U) \setminus f^{-1}(V)$. Hence, $f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$. □

**Exercise 3.4.5**. *Let $f : X \to Y$ be a function from one set $X$ to another set $Y$. Show that $f(f^{-1}(S)) = S$ for every $S \subseteq Y$ if and only if $f$ is surjective. Show that $f^{-1}(f(S))$ for every $S \subseteq X$ if and only if $f$ is injective.*

*Solution.*

(a) Show that $f(f^{-1}(S)) = S$ for every $S \subseteq Y$ if and only if $f$ is surjective.

We will first prove the implication. Suppose $f(f^{-1}(S)) = S$ for every $S \subseteq Y$; we need to show that for all $y \in Y$, there exists an $x \in X$ such that $f(x) = y$. Consider the subset $S = Y$ which contains all elements of $Y$. By the initial assumption, we have $Y = f(f^{-1}(Y))$, thus $y \in f(f^{-1}(Y))$ for any $y \in Y$. From the definition of forward images, there exists an $x \in f^{-1}(Y)$ such that $f(x) = y$, which is exactly our claim.

Now consider the converse. Suppose $f$ is surjective; we need to show that $f(f^{-1}(S)) = S$. Let $y$ be an arbitrary element of $S$. Because $f$ is surjective, there exists an $x \in X$ such that $f(x) = y$. Thus, $f(x) \in S$, which implies $x \in f^{-1}(S)$ by the definition of inverse images. From the definition of forward images, $f(x) \in f(f^{-1}(S))$, which leads to $y \in f(f^{-1}(S))$. From Exercise 3.4.2, we know $f(f^{-1}(S)) \subseteq S$. Hence, $f(f^{-1}(S)) = S$ if $f$ is surjective.

As the implication and its converse are both true, we have $f(f^{-1}(S)) = S$ for every $S \subseteq Y$ if and only if $f$ is surjective.

(b) Show that $f^{-1}(f(S)) = S$ for every $S \subseteq X$ if and only if $f$ is injective.

We will first prove the implication. Suppose $f^{-1}(f(S)) = S$ for every $S \subseteq X$; we need to show that $f$ is injective. Let $x, y \in X$ such that $f(x) = f(y)$. We want to show that $x = y$. Take $S = \{x\}$. Then, $f(\{x\}) = \{f(x)\}$. Since $f(x) = f(y)$, $f(y)$ is an element of $f(\{x\})$, therefore $y \in f^{-1}(f(\{x\}))$ by the definition of inverse images. Also, by the hypothesis, we have $f^{-1}(f(\{x\})) = \{x\}$. Therefore, we have $y \in \{x\}$, which implies $y = x$. Hence, we can conclude that if $f^{-1}(f(S)) = S$ for every $S \subseteq X$, then $f$ is injective.

Now consider the converse. Suppose $f$ is injective; we need to show $f^{-1}(f(S)) = S$ for every $S \subseteq X$. Let $x \in f^{-1}(f(S))$. Then, by definition, we have $f(x) \in f(S)$, and thus there exists some $x' \in S$ such that $f(x) = f(x')$. Because $f$ is injective, we have $x = x'$, thus $x \in S$. Thus, $f^{-1}(f(S)) \subseteq S$. From Exercise 3.4.2, we know $S \subseteq f^{-1}(f(S))$. Hence, we can conclude that if $f$ is injective, then $f^{-1}(f(S)) = S$ for every $S \subseteq X$.

Because the implication and its converse are both true, we have $f^{-1}(f(S)) = S$ for every $S \subseteq X$ if and only if $f$ is injective. □

**Exercise 3.4.6**. *Prove Lemma 3.4.9: Let $X$ be a set. Then the set*

$$\{Y : Y \textit{ is a subset of } X\}$$

*is a set.*

*Solution.* By the power set axiom, $\{0,1\}^X$ is a set, and its elements are exactly the functions mapping from $X$ to $\{0,1\}$. For each function $f \in \{0,1\}^X$, define $A_f \coloneqq f^{-1}(\{1\})$. Since $f$ has a domain of $X$, every element of $A_f$ is an element of $X$, thus we have $A_f \subseteq X$.

Let $A$ be an arbitrary subset of $X$. Then, we can define the function $f_A : X \to \{0,1\}$ such that for all $x \in X$, we have $f_A(x) = 1$ if $x \in A$, and $f_A(x) = 0$ otherwise. Note that $f_A^{-1}(\{1\}) = A$. Also, because $f_A$ has domain $X$ and range $\{0,1\}$, we have $f_A \in \{0,1\}^X$. Thus, $A$ is a subset of $X$ if and only if $A = f^{-1}(X)$ for some $f \in \{0,1\}^X$. For each $f$, there is exactly one $A$ such that $f^{-1}(\{1\}) = A$. Hence, by the axiom of replacement, the collection $\{A : f^{-1}(\{1\})$ for some $f \in \{0,1\}^X\}$ exists. This is equivalent to saying $\{A : A$ is a subset of $X\}$. □

**Exercise 3.4.7**. *Let $X$, $Y$ be sets. Define a* partial function *from $X$ to $Y$ to be any function $f : X' \to Y'$ whose domain $X'$ is a subset of $X$, and whose range $Y'$ is a subset of $Y$. Show that the collection of all partial functions from $X$ to $Y$ is itself a set.*

*Solution.* From the problem statement, we have $X' \subseteq X$ and $Y' \subseteq Y$. For any $X'$ and $Y'$, we have $Y'^{X'}$, a set with all functions mapping from $X'$ to $Y'$. Also, by Lemma 3.4.9, we have the set $2^X$, the set of all subsets of $X$, and $2^Y$, the set of all subsets of $Y$. Thus, $X' \in 2^X$ and $Y' \in 2^Y$.

We first fix a certain $X'$. Let $A$ be a set, and $P(Y', A)$ be the property pertaining to $Y'$ that $A = Y'^{X'}$. Then, for any $Y' \in 2^Y$, $P(Y', A)$ will be true for exactly one $A$. Therefore, by the replacement axiom, the set $\{A : A = Y'^{X'} \text{ for some } Y' \in 2^Y\} = \{Y'^{X'} : Y' \in 2^Y\}$ exists. Because each of the elements of this set is a set, we also have the set $\bigcup\{Y'^{X'} : Y' \in 2^Y\} = \{f \mid f : X' \to Y' \text{ for some } Y' \in 2^Y\}$. Write this set as $S_{X'}$.

Now consider $X' \in 2^X$. For every $X'$, we have exactly one set $S_{X'}$. According to the replacement axiom (Axiom 3.6), the set $\{S_{X'} : X' \in 2^X\}$ exists. By the union axiom (Axiom 3.11), this means the set $\bigcup_{X' \in 2^X} S_{X'}$ exists. For any function $f$, $f$ is an element of this set if and only if we have $f : X' \to Y'$ for some $X' \subseteq X$ and $Y' \subseteq Y$. This is precisely the collection of all partial functions from $X$ to $Y$. □

**Exercise 3.4.8**. *Show that Axiom 3.4 can be deduced from Axiom 3.1, Axiom 3.3 and Axiom 3.11.*

*Solution.* Let $A$ and $B$ be sets. According to Axiom 3.1, the sets $A$ and $B$ are objects, and therefore can be elements of sets. Therefore, we can construct the pair set $\{A, B\}$. From Axiom 3.11, there exists a set $\bigcup\{A, B\}$ such that for any $x \in \bigcup\{A, B\}$, $x \in S$ for some $S \in \{A, B\}$. However, since $\{A, B\}$ is a pair set with two elements, we have $S = A$ or $S = B$. Hence, $x \in A$ or $x \in B$ if and only if $x \in \bigcup\{A, B\}$. We denote $\bigcup\{A, B\} = A \cup B$; this is precisely the definition of pairwise union as in Axiom 3.4. □

**Exercise 3.4.9**. *Show that if $\beta$ and $\beta'$ are two elements of a set $I$, and to each $\alpha \in I$ we assign a set $A_\alpha$, then*

$$\{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\} = \{x \in A_{\beta'} : x \in A_\alpha \text{ for all } \alpha \in I\},$$

*and so the definition of $\bigcap_{\alpha \in I} A_\alpha$ defined in (3.3) does not depend on $\beta$. Also explain why (3.4) is true.*

*Solution.* Let $x$ be an element of $\{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\}$. Then, by the axiom of specification, $x$ is an element of $A_\alpha$ for all $\alpha \in I$. This implies $x$ is an element of $A_{\beta'}$, and thus $\{x \in A_{\beta'} : x \in A_\alpha \text{ for all } \alpha \in I\}$. The same argument can be made with the role of $\beta$ and $\beta'$ inverted. Hence, we have $\{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\} = \{x \in A_{\beta'} : x \in A_\alpha \text{ for all } \alpha \in I\}$ as desired. Applying the specification axiom, we have $x \in \bigcap_{\alpha\in I} A_\alpha$ if and only if $x \in A_\alpha$ for all $\alpha \in I$, which is exactly as stated in (3.4). $\square$

**Exercise 3.4.10**. *Suppose that $I$ and $J$ are two sets, and for all $\alpha \in I \cup J$ let $A_\alpha$ be a set. Show that $(\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha) = \bigcup_{\alpha\in I\cup J} A_\alpha$. If $I$ and $J$ are non-empty, show that $(\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha) = \bigcap_{\alpha\in I\cup J} A_\alpha$.*

*Solution.*

(a) Show that $(\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha) = \bigcup_{\alpha\in I\cup J} A_\alpha$.

We will first consider the left-to-right implication. Let $y \in (\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha)$. Then, $y \in \bigcup_{\alpha\in I} A_\alpha$ or $y \in \bigcup_{\alpha\in J} A_\alpha$. If $y \in \bigcup_{\alpha\in I} A_\alpha$, then $y \in A_\alpha$ for some $\alpha \in I$. This implies $y \in A_\alpha$ for some $\alpha \in I \cup J$, or $y \in \bigcup_{\alpha\in I\cup J} A_\alpha$. Using a similar argument, if $y \in \bigcup_{\alpha\in J} A_\alpha$, then we have $y \in \bigcup_{\alpha\in I\cup J} A_\alpha$. Thus, we have $(\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha) \subseteq \bigcup_{\alpha\in I\cup J} A_\alpha$.

Now consider the converse. Let $y \in \bigcup_{\alpha\in I\cup J} A_\alpha$. Then, $y \in A_\alpha$ for some $\alpha \in I \cup J$. We will split into two cases. If $y \in A_\alpha$ for some $\alpha \in I$, then $y \in \bigcup_{\alpha\in I} A_\alpha$. Similarly, if $y \in A_\alpha$ for some $\alpha \in J$, then $y \in \bigcup_{\alpha\in J} A_\alpha$. Therefore, we have $y \in (\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha)$. This is the same as $\bigcup_{\alpha\in I\cup J} A_\alpha \subseteq (\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha)$.

Therefore, we have $(\bigcup_{\alpha\in I} A_\alpha) \cup (\bigcup_{\alpha\in J} A_\alpha) = \bigcup_{\alpha\in I\cup J} A_\alpha$.

(b) If $I$ and $J$ are non-empty, show that $(\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha) = \bigcap_{\alpha\in I\cup J} A_\alpha$.

We will first consider the left-to-right implication. Let $y \in (\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha)$. Then, $y \in \bigcap_{\alpha\in I} A_\alpha$ and $y \in \bigcap_{\alpha\in J} A_\alpha$. This means $y \in A_\alpha$ for all $\alpha \in I$ and for all $\alpha \in J$. Thus, we have $y \in A_\alpha$ for all $\alpha \in I \cup J$, or $y \in \bigcap_{\alpha\in I\cup J} A_\alpha$. Hence, $(\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha) \subseteq \bigcap_{\alpha\in I\cup J} A_\alpha$.

Now consider the converse. Let $y \in \bigcap_{\alpha\in I\cup J} A_\alpha$. Then, $y \in A_\alpha$ for all $\alpha \in (I \cup J)$. This leads to $y \in A_\alpha$ for all $\alpha \in I$ and for all $\alpha \in J$. Therefore, we have $y \in \bigcap_{\alpha\in I} A_\alpha$ and $y \in \bigcap_{\alpha\in J} A_\alpha$. We then have $y \in (\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha)$. Hence, $\bigcap_{\alpha\in I\cup J} A_\alpha \subseteq (\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha)$.

Therefore, we have $(\bigcap_{\alpha\in I} A_\alpha) \cap (\bigcap_{\alpha\in J} A_\alpha) = \bigcap_{\alpha\in I\cup J} A_\alpha$ as required. $\square$

**Exercise 3.4.11**. *Let $X$ be a set, let $I$ be a non-empty set, and for all $\alpha \in I$ let $A_\alpha$ be a subset of $X$. Show that*

$$X \setminus \bigcup_{\alpha\in I} A_\alpha = \bigcap_{\alpha\in I}(X \setminus A_\alpha) \text{ and } X \setminus \bigcap_{\alpha\in I} A_\alpha = \bigcup_{\alpha\in I}(X \setminus A_\alpha).$$

*Solution.* We will prove the first equation. Let $x \in X \setminus \bigcup_{\alpha\in I} A_\alpha$. Then, we have

$$\begin{aligned} x \in X \setminus \bigcup_{\alpha\in I} A_\alpha &\iff x \in X \text{ and } x \notin \bigcup_{\alpha\in I} A_\alpha \\ &\iff x \in X \text{ and } x \notin A_\alpha \text{ for any } \alpha \in I \\ &\iff x \in X \setminus A_\alpha \text{ for any } \alpha \in I \\ &\iff x \in \bigcap_{\alpha\in I}(X \setminus A_\alpha) \end{aligned}$$

Thus, $X \setminus \bigcup_{\alpha\in I} A_\alpha = \bigcap_{\alpha\in I}(X \setminus A_\alpha)$.

We will now prove the second equation in a similar manner. Let $x \in X \setminus \bigcap_{\alpha\in I} A_\alpha$. Then, we have

$$\begin{aligned} x \in X \setminus \bigcap_{\alpha\in I} A_\alpha &\iff x \in X \text{ and } x \notin \bigcap_{\alpha\in I} A_\alpha \\ &\iff x \in X \text{ and } x \notin A_\alpha \text{ for some } \alpha \in I \\ &\iff x \in X \setminus A_\alpha \text{ for some } \alpha \in I \\ &\iff x \in \bigcup_{\alpha\in I}(X \setminus A_\alpha) \end{aligned}$$

Thus, $X \setminus \bigcap_{\alpha\in I} A_\alpha = \bigcup_{\alpha\in I}(X \setminus A_\alpha)$ as required. □

## 3.5 Cartesian products

**Exercise 3.5.1**. *Suppose we define the ordered pair $(x, y)$ for any objects $x$ and $y$ by the formula $(x, y) := \{\{x\}, \{x, y\}\}$. Show that such a definition indeed obeys the property (3.5), and also whenever $X$ and $Y$ are sets, the Cartesian product $X \times Y$ is also a set. Thus this definition can be validly used as a definition of an ordered pair. For an additional challenge, show that the alternate definition $(x, y) := \{x, \{x, y\}\}$ also verifies (3.5) and is thus also an acceptable definition of ordered pair.*

*Solution.* Recall that (3.5) states two ordered pairs $(x, y)$ and $(x', y')$ are equal if and only if $x = x'$ and $y = y'$. Write $(x, y) = \{\{x\}, \{x, y\}\}$, $(x', y') = \{\{x'\}, \{x', y'\}\}$.

(a) Show that such a definition indeed obeys the property (3.5).

The right-to-left implication is easy to prove: if $x = x'$ and $y = y'$, then $\{\{x\}, \{x, y\}\} = \{\{x'\}, \{x', y'\}\}$ by the substitution axiom (see Appendix A.7), thus $(x, y) = (x', y')$. Now consider the converse; we want to show if $\{\{x\}, \{x, y\}\} = \{\{x'\}, \{x', y'\}\}$, then $x = x'$ and $y = y'$. We will consider two cases.

If $x = y$, then $\{\{x\}, \{x, y\}\}$ simplifies to the singleton $\{\{x\}\}$. Then, we have $\{\{x'\}, \{x', y'\}\} = \{\{x\}\}$. Therefore, $\{\{x'\}, \{x', y'\}\}$ is also a singleton. A pair set is a singleton if and only if its elements are equal, therefore $\{x'\} = \{x', y'\}$, and thus $x' = y'$. We can also deduce $\{x\} \in \{\{x'\}, \{x', y'\}\}$, which leads to $x = x'$. By the hypothesis, we have $x = y$, thus we also have $y = y'$ because $x' = y'$, thus the property is true in the case that $x = y$.

Consider instead the case where $x \neq y$. By the definition of equality of sets, we have $\{x\} \in \{\{x'\}, \{x', y'\}\}$, which implies $\{x\} = \{x'\}$ or $\{x\} = \{x', y'\}$. The latter is false as the left-hand side is a singleton while the right-hand side is a pair set. Thus, we have $\{x\} = \{x'\}$, or $x = x'$. Similarly, we have $\{x, y\} \in \{\{x'\}, \{x', y'\}\}$, which implies $\{x, y\} = \{x', y'\}$ as $\{x, y\} \neq \{x\}$. Therefore, $y \in \{x', y'\}$, which leads to $y = x'$ or $y = y'$. Because $x = x'$ and $y \neq x$, we have $y \neq x'$, thus $y = y'$. We can now conclude that the property is true when $x \neq y$.

(b) Show that whenever $X$ and $Y$ are sets, the Cartesian product $X \times Y$ is also a set.

Because $x \in X$ and $y \in Y$, $\{x\}$ and $\{x, y\}$ are subsets of $X \cup Y$. Therefore, both $\{x\}$ and $\{x, y\}$ are elements of $\mathcal{P}(X \cup Y)$. We can then construct the set $\{\{x\}, \{x, y\}\}$, which is a subset of $\mathcal{P}(X \cup Y)$, and thus an element of $\mathcal{P}(\mathcal{P}(X \cup Y))$. By the axiom of specification, we can construct a set $\{S \in \mathcal{P}(\mathcal{P}(X \cup Y)) : S = \{\{x\}, \{x, y\}\}$ for some $x \in X, y \in Y\}$. Therefore, if an object $a$ is in this set, $a$ is an ordered set $\{\{x\}, \{x, y\}\}$ for some $x \in X$ and $y \in Y$. This is equivalent to the definition of the Cartesian product, $X \times Y$.

(c) Show that the alternate definition $(x, y) \coloneqq \{x, \{x, y\}\}$ also verifies (3.5).

With the alternate definition, we want to show

$$\{x, \{x, y\}\} = \{x', \{x', y'\}\} \iff x = x' \text{ and } y = y'.$$

It is easy to prove the right-to-left implication: suppose $x = x'$ and $y = y'$; then $\{x, \{x, y\}\} = \{x', \{x', y'\}\}$ by the substitution axiom, thus $(x, y) = (x', y')$.

Now consider the converse. Suppose $\{x, \{x, y\}\} = \{x', \{x', y'\}\}$; we want to show $x = x'$ and $y = y'$. By the definition of equality of sets, we have $x \in \{x', \{x', y'\}\}$, thus $x = x'$ or $x = \{x', y'\}$. Similarly, $x' \in \{x, \{x, y\}\}$, thus $x = x'$ or $x' = \{x, y\}$. Suppose $x \neq x'$, thus $x = \{x', y'\}$ and $x' = \{x, y\}$. This means $x$ and $x'$ are members of each other: a direct contradiction to Exercise 3.2.2. Thus, we have $x = x'$. As $x = x'$ and $\{x, y\} \neq x$ due to the axiom of regularity, we have $\{x, y\} = \{x', y'\}$. Assume $y = y'$. As $y \in \{x', y'\}$ and $y' \in \{x, y\}$, we have $y = x'$ and $y' = x$. Because $x = x'$, we have $y = y'$, a contradiction. Therefore, $y = y'$. Hence, the property (3.5) is valid with this definition of ordered sets.

*Remark.* My initial thought for (b) was to construct the set $X \times Y$ by the replacement axiom instead of looking for a master set. However, it is less connected with the given definitions of ordered sets. Nevertheless, I present the alternative solution:

We first fix an arbitrary value of $y \in Y$, and let $x \in X$. Let $A$ be an arbitrary set. Because $y$ is fixed, there is exactly one $A$ such that $A = \{x, \{x, y\}\}$ by uniqueness of sets. Therefore, by the axiom of replacement, the set $\{A : A = \{x, \{x, y\}\} \text{ for some } x \in X\}$ exists. We simplify this set as $\{\{x, \{x, y\}\} \text{ for some } x \in X\}$, and denote this set as $A_y$.

Because for every element $y \in Y$ we have some set $A_y$, we can form the union set $\bigcup_{y \in Y} A_y = \bigcup\{A_y : y \in Y\} = \{\{x, \{x, y\}\} \text{ for some } x \in X, y \in Y\}$.

**Exercise 3.5.2**. *Suppose we define an ordered $n$-tuple to be a surjective function $x : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X$ whose range is some arbitrary set $X$ (so different ordered $n$-tuples are allowed to have different ranges); we then write $x_i$ for $x(i)$, and also write $x$ as $(x_i)_{1 \leq i \leq n}$. Using this definition, verify that we have $(x_i)_{1 \leq i \leq n} = (y_i)_{1 \leq i \leq n}$ if and only if $x_i = y_i$ for all $1 \leq i \leq n$. Also, show that if $(X_i)_{1 \leq i \leq n}$ is an ordered $n$-tuple of sets, then the Cartesian product, as defined in Definition 3.5.7, is indeed a set.*

*Solution.* For the first statement, we have

$$\begin{aligned}(x_i)_{1\leq i\leq n} = (y_i)_{1\leq i\leq n} &\iff x(i) = y(i) \text{ for all } 1 \leq i \leq n \\ &\iff x_i = y_i \text{ for all } 1 \leq i \leq n.\end{aligned}$$

Because ordered $n$-tuples are defined to be surjective, $x$ and $y$ have the same range.

For the second statement, suppose $(X_i)_{1\leq i\leq n}$ is an ordered $n$-tuple of sets. Let $I = \{i \in \mathbb{N} : 1 \leq i \leq n\}$. From the union axiom (Axiom 3.11), the set $A = \bigcup\{X_i : 1 \leq i \leq n\}$ contains all elements from $X_1$ to $X_n$. From Exercise 3.4.7, the collection of all partial functions from $I$ to $A$ is a set. This set certainly contains all functions that has a domain of $I$ and range a subset of $A$. Using the axiom of specification, we can therefore produce the set $\{x : I \to A : x(i) \in X_i \text{ for all } 1 \leq i \leq n\}$. This is equivalent to the set $\{(x_i)_{1\leq i\leq n} : x(i) \in X_i \text{ for all } 1 \leq i \leq n\}$, the definition as in Definition 3.5.7. □

**Exercise 3.5.3**. *Show that the definition of equality for ordered pair and ordered $n$-tuple obey the reflexivity, symmetry and transitivity axioms.*

*Solution.* We first show such properties for ordered pairs.

(a) *Reflexivity: $(x, y) = (x, y)$.*

We have $x = x$ and $y = y$, which is true by reflexivity of objects.

(b) *Symmetry: If $(x, y) = (x', y')$, then $(x', y') = (x, y)$.*

We have $x = x'$ and $y = y'$ by the definition of equality of ordered tuples (Definition 3.5.1), thus $x' = x$, $y' = y$ by symmetry of objects. Therefore, applying Definition 3.5.1 again, we have $(x', y') = (x, y)$.

(c) *Transitivity: If $(x, y) = (x', y')$ and $(x', y') = (x'', y'')$, then $(x, y) = (x'', y'')$.*

The first equality gives $x = x'$, $y = y'$, while the second equality gives $x' = x''$ and $y' = y''$. By the transitivity of objects, we also have $x = x''$ and $y = y''$. Hence, using Definition 3.5.1, we have $(x, y) = (x'', y'')$.

We will now show such properties for ordered $n$-tuples.

(a) *Reflexivity: $(x_i)_{1\leq i\leq n} = (x_i)_{1\leq i\leq n}$*

By the definition of equality of ordered tuples (Definition 3.5.7), we have $x_i = x_i$ for all $1 \leq i \leq n$, which is true by reflexivity of objects.

(b) *Symmetry: If* $(x_i)_{1\leq i\leq n} = (y_i)_{1\leq i\leq n}$, *then* $(y_i)_{1\leq i\leq n} = (x_i)_{1\leq i\leq n}$.

For all $1 \leq i \leq n$, we have $x_i = y_i$ by definition of equality of ordered tuples (Definition 3.5.7). Thus, we have $y_i = x_i$ for all $1 \leq i \leq n$ due to the symmetry of objects. Applying Definition 3.5.7 again, we have $(y_i)_{1\leq i\leq n} = (x_i)_{1\leq i\leq n}$.

(c) *Transitivity: If* $(x_i)_{1\leq i\leq n} = (y_i)_{1\leq i\leq n}$ *and* $(y_i)_{1\leq i\leq n} = (z_i)_{1\leq i\leq n}$, *then* $(x_i)_{1\leq i\leq n} = (z_i)_{1\leq i\leq n}$.

According to the definition of equality of sets (Definition 3.5.7), we have $x_i = y_i$ and $y_i = z_i$ for all $1 \leq i \leq n$. Thus, by the transitivity of objects, we have $x_i = z_i$ for all $1 \leq i \leq n$. Using Definition 3.5.7 again, we obtain $(x_i)_{1\leq i\leq n} = (z_i)_{1\leq i\leq n}$. $\square$

**Exercise 3.5.4**. *Let $A$, $B$, $C$ be sets. Show that $A \times (B \times C) = (A \times B) \cup (A \times C)$, that $A \times (B \cap C) = (A \times B) \cap (A \times C)$, and that $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.*

*Solution.* First, we prove a simple lemma:

**Lemma.** *An ordered set $(x, y)$ is an element of the Cartesian product $X \times Y$ if and only if $x \in X$ and $y \in Y$. In other words,*

$$(x, y) \in X \times Y \iff x \in X \text{ and } y \in Y.$$

*Proof.* By definition, $X \times Y$ is the set of all ordered pairs $(x', y')$ for some $x' \in X$ and $y' \in Y$. Therefore, an ordered pair $(x, y)$ is an element of this set if and only if $x \in X$ and $y \in Y$. $\square$

We want to show if an ordered set $(x, y)$ is in either one of the set if and only if it is also in the other.

(a) Show that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

$$\begin{aligned}(x,y) \in A \times (B \cup C) &\iff x \in A \text{ and } y \in B \cup C \\ &\iff x \in A \text{ and } (y \in B \text{ or } y \in C) \\ &\iff (x \in A \text{ and } y \in B) \text{ or } (x \in A \text{ and } y \in C) \\ &\iff ((x,y) \in A \times B) \text{ or } ((x,y) \in A \times C) \\ &\iff (x,y) \in (A \times B) \cup (A \times C).\end{aligned}$$

(b) Show that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

$$\begin{aligned}(x,y) \in A \times (B \cap C) &\iff x \in A \text{ and } y \in B \cap C \\ &\iff x \in A \text{ and } (y \in B \text{ and } y \in C) \\ &\iff (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \in C) \\ &\iff (x,y) \in A \times B \text{ and } (x,y) \in A \times C \\ &\iff (x,y) \in (A \times B) \cap (A \times C)\end{aligned}$$

(c) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

$$\begin{aligned}(x,y) \in A \times (B \setminus C) &\iff x \in A \text{ and } y \in B \setminus C \\ &\iff x \in A \text{ and } (y \in B \text{ and } y \notin C) \\ &\iff (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \notin C) \\ &\iff (x,y) \in A \times B \text{ and } (x,y) \notin A \times C \\ &\iff (x,y) \in (A \times B) \setminus (A \times C) \quad \square\end{aligned}$$

**Exercise 3.5.5**. *Let $A$, $B$, $C$, $D$ be sets. Show that $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$. Is it true that $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$? Is it true that $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$?*

*Solution.*

(a) Show that $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

$$\begin{aligned}(x,y) \in (A \times B) \cap (C \times D) &\iff ((x,y) \in A \times B) \text{ and } ((x,y) \in C \times D) \\ &\iff (x \in A \text{ and } y \in B) \text{ and } (x \in C \text{ and } y \in D) \\ &\iff (x \in A \text{ and } x \in C) \text{ and } (y \in B \text{ and } y \in D) \\ &\iff (x \in A \cap C) \text{ and } (y \in B \cap D) \\ &\iff (x,y) \in (A \cap C) \times (B \cap D).\end{aligned}$$

(b) Is it true that $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$?

This statement is false. Let $A = \{1\}$, $B = \{2\}$, $C = \{3\}$, $D = \{4\}$. Then, $(A \times B) \cup (C \times D) = \{(1,2),(3,4)\}$, but $(A \cup C) \times (B \cup D) = \{(1,2),(1,4),(3,2),(3,4)\}$.

(c) Is it true that $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$?

This statement is also false. Let $A = \{1,2\}$, $B = \{3,4\}$, $C = \{2\}$, $D = \{4\}$. Then $(A \times B) \setminus (C \times D) = \{(1,3),(1,4),(2,3)\}$, but $(A \setminus C) \times (B \setminus D) = \{(1,3)\}$.

**Exercise 3.5.6**. *Let $A$, $B$, $C$, $D$ be non-empty sets. Show that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq C$, and that $A \times B = C \times D$ if and only if $A = C$ and $B = D$. What happens if the hypotheses that $A$, $B$, $C$, $D$ are all non-empty are removed?*

*Solution.*

(a) Show that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq D$.

Let $A$, $B$, $C$, $D$ be non-empty sets. We will first prove the left-to-right implication. Suppose $a \in A$, $b \in B$. Since $A \times B \subseteq C \times D$, we have $(a,b) \in C \times D$. By the lemma from the previous exercise, $a \in C$ and $b \in D$. Hence, $A \subseteq C$ and $B \subseteq D$.

Now consider the converse. Suppose $A \subseteq C$ and $B \subseteq D$. Let $(a,b) \in A \times B$. Then, $a \in A$ and $b \in B$. Since $A \subseteq C$, $a \in C$. Since $B \subseteq D$, $b \in D$. Thus, we have $(a,b) \in C \times D$. This implies $A \times B \subseteq C \times D$.

(b) Show that $A \times B = C \times D$ if and only if $A = C$ and $B = D$.

We will first prove the left-to-right implication. Suppose $A \times B = C \times D$; then we have $A \times B \subseteq C \times D$. From the first part of this exercise, this implies $A \subseteq C$ and $B \subseteq D$. Similarly, $C \times D \subseteq A \times B$ implies $C \subseteq A$ and $D \subseteq B$. Hence, we have $A \times B = C \times D$ if $A = C$ and $B = D$.

Now consider the converse. Suppose $A = C$ and $B = D$. Then, we have $A \subseteq C$ and $B \subseteq D$. From the first part of this exercise, this implies $A \times B \subseteq C \times D$. Similarly, $C \subseteq D$ and $B \subseteq A$ implies $C \times D \subseteq A \times B$. Hence, $A = C$ and $B = D$ if $A \times B = C \times D$.

(c) What happens if the hypotheses that $A$, $B$, $C$, $D$ are all non-empty are removed?

The statements would then be false. Let $A = \{0\}$, $B = C = D = \varnothing$. Then, $A \times B = C \times D = \varnothing$, but $A \not\subseteq C$. □

**Exercise 3.5.7**. *Let $X$, $Y$ be sets, and let $\pi_{X\times Y\to X} : X \times Y \to X$ and $\pi_{X\times Y\to Y} : X \times Y \to Y$ be the maps $\pi_{X\times Y\to X}(x, y) := x$ and $\pi_{X\times Y\to Y}(x, y) := y$; these maps are known as the* co-ordinate functions *on $X \times Y$. Show that for any functions $f : Z \to X$ and $g : Z \to Y$ , there exists a unique function $h : Z \to X \times Y$ such that $\pi_{X\times Y\to X} \circ h = f$ and $\pi_{X\times Y\to Y} \circ h = g$. This function $h$ is known as the* direct sum *of $f$ and $g$ and is denoted $h = f \oplus g$.*

*Solution.* We will first show the existence of the function $h$. Define $h$ as such: $h(z) = (f(z), g(z))$ for any $z \in Z$. Thus, the range of $h$ is $X \times Y$ Thus, we have the following equalities:

$$(\pi_{X\times Y\to X} \circ h)(z) = \pi_{X\times Y\to X}(h(x)) = \pi_{X\times Y\to X}((f(x), g(x)) = f(x)$$

$$(\pi_{X\times Y\to Y} \circ h)(z) = \pi_{X\times Y\to Y}(h(x)) = \pi_{X\times Y\to Y}((f(x), g(x)) = g(x)$$

Therefore, for this definition of $h$, the condition holds.

We will now show the uniqueness of $h$. Suppose there is another function $h'$ such that $\pi_{X\times Y\to X} \circ h = f$ and $\pi_{X\times Y\to Y} \circ h = g$. For any $z \in Z$, $h'(z)$ must be an ordered set $(a, b)$. Then, we have

$$a = (\pi_{X\times Y\to X} \circ h')(z) = f(z) \qquad b = (\pi_{X\times Y\to Y} \circ h')(z) = g(z)$$

Thus, $h'(z) = (f(z), g(z))$ for all $z \in Z$, which is exactly equal as our initial definition. Hence, the function $h$ is unique. □

**Exercise 3.5.8**. *Let $X_1, \ldots, X_n$ be sets. Show that the Cartesian product $\prod_{i=1}^n X_i$ is empty if and only if at least one of the $X_i$ is empty.*

*Solution.* We will first prove the left-to-right implication: If at least one of the $X_i$ is empty, then the Cartesian product $\prod_{i=1}^n X_i$ is empty. Suppose at least one of the $X_i$ is empty. Then, all objects are not elements of that set. From the definition of Cartesian products, this implies there is no ordered set $(x_i)_{1\leq i\leq n}$ such that $x_i \in X_i$ for all $1 \leq i \leq n$. As such, the Cartesian product is empty.

Now consider the right-to-left implication: If the Cartesian product $\prod_{i=1}^{n} X_i$ is empty, then at least one of the $X_i$ is empty. If all $X_i$ are non-empty, then the Cartesian product will not be empty as per Lemma 3.5.12. Thus, if the Cartesian product is empty, at least one of the $X_i$ is empty. □

**Exercise 3.5.9**. *Suppose that $I$ and $J$ are two sets, and for all $\alpha \in I$ let $A_\alpha$ be a set, and for all $\beta \in J$ let $B_\beta$ be a set. Show that $(\bigcup_{\alpha \in I} A_\alpha) \cap (\bigcup_{\beta \in J} B_\beta) = \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$.*

*Solution.* We will first prove the left-to-right implication. Let $x \in (\bigcup_{\alpha \in I} A_\alpha) \cap (\bigcup_{\beta \in J} B_\beta)$. Then, $x \in \bigcup_{\alpha \in I} A_\alpha$ and $x \in \bigcup_{\beta \in J} B_\beta$. We have $x \in A_\alpha$ for some $\alpha \in I$ and $x \in B_\beta$ for some $\beta \in J$. Therefore, $x \in A_\alpha \cap B_\beta$ for some $\alpha \in I$ and $\beta \in J$. This is the same as saying $x \in A_\alpha \cap B_\beta$ for some $(\alpha, \beta) \in I \times J$. Hence, $x \in \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$.

Now consider the converse. Let $x \in \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$. We have $x \in A_\alpha \cap B_\beta$ for some $(\alpha, \beta) \in I \times J$. Then, it follows that $x \in A_\alpha$ and $x \in \beta$ for some $\alpha \in I$ and $\beta \in J$. We can rewrite this as $x \in \alpha$ for some $\alpha in I$ and $x \in \beta$ for some $\beta \in J$. Thus, $x \in \bigcup_{\alpha \in I}$ and $x \in \bigcup_{\beta \in J}$, or $x \in (\bigcup_{\alpha \in I}) \cap (\bigcup_{\beta \in J})$.

**Exercise 3.5.10**. *If $f : X \to Y$ is a function, define the* graph *of $f$ to be the subset of $X \times Y$ defined by $\{(x, f(x)) : x \in X\}$. Show that two functions $f : X \to Y$ , $\tilde{f} : X \to Y$ are equal if and only if they have the same graph. Conversely, if $G$ is any subset of $X \times Y$ with the property that for each $x \in X$, the set $\{y \in Y : (x, y) \in G\}$ has exactly one element (or in other words, $G$ obeys the* vertical line test*), show that there is exactly one function $f : X \to Y$ whose graph is equal to $G$.*

*Solution.* We will first show the first part of the exercise. If $f = \tilde{f}$, then for any $x \in X$, $f(x) = \tilde{f}(x)$, and thus $(x, f(x)) = (x, \tilde{f}(x))$ for all $x$. Then the graphs of $f$ and $\tilde{f}$ are equal. Conversely, if the graphs of $f$ and $\tilde{f}$ are equal, then $(x, f(x)) = (x, \tilde{f}(x))$ for all $x \in X$. This implies $f(x) = \tilde{f}(x)$ for all $x$; it follows that $f = \tilde{f}$.

Now consider the second part of the exercise. We will first show the existence of the function $f$. Denote the set $\{y \in Y : (x, y) \in G\}$ as $Y_x$. For each $x$, there is exactly one $y \in Y$ such that $(x, y) \in G$. Denote this $y$ as $y_x$. We can define $f : X \to Y$ such that $f(x) = y_x$. This is a valid function because for each $x \in X$, there is exactly one value of $y \in Y$. The graph of $f$ is $\{(x, f(x)) : x \in X\}$. If $(x, y) \in G$, then $y \in Y_x$. As $Y_x$ has exactly one element, it follows that $y = y_x = f(x)$. Thus, $(x, y) = (x, f(x))$, which is in the graph of $f$. Conversely, if $(x, y)$ is an element in the graph of $f$, then

$y = f(x) = y_x$, and thus $y \in Y_x$, $(x, y) \in G$. Hence, the graph of $f$ is equal to $G$.

We will now show the uniqueness of the function $f$. Suppose there is another function $h$ whose graph is equal to $G$. Then, the graph of $h$ is equal to the graph of $f$. From the first part of the exercise, this implies $h$ and $f$ are equal functions. Hence, the function $f$ is unique. $\square$

**Exercise 3.5.11**. *Show that Axiom 3.10 can in fact be deduced from Lemma 3.4.9 and the other axioms of set theory, and thus Lemma 3.4.9 can be used as an alternate formulation of the power set axiom.*

*Solution.* By the definition of Cartesian products (Definition 3.5.7), the Cartesian product $X \times Y$ is a set. From Lemma 3.4.9, the collection of all subsets of $X \times Y$ is also a set ($\mathcal{P}(X \times Y)$. Using the axiom of specification (Axiom 3.5), we can construct the following set:

$$A = \{G \in \mathcal{P}(X \times Y) : G \text{ obeys the vertical line test}\}.$$

Note that $A \subseteq \mathcal{P}(X \times Y)$. According to Exercise 3.5.10, each element in $A$ is a graph. For each $G \in A$, there is exactly one $f$ whose graph is exactly equal to $G$. By the axiom of replacement (Axiom 3.6), we can now construct the following set:

$$\{f : f \text{ has the graph } G \text{ for some } G \in \mathcal{P}(X \times Y)\}$$

which is exactly equal to $Y^X$, the set of all functions with domain $X$ and range $Y$. This is exactly the definition of $Y^X$ in Axiom 3.10 (power set axiom). $\square$

*(Exercise 3.5.12 and Exercise 3.5.13 are skipped for the time being.)*